



Investigation on fatigue life prediction of engineering welded structures: A physics-informed data-driven approach



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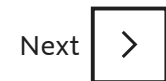
Highlights

- Dual-driven DCNN embeds objective fatigue laws as dynamic loss function penalty constraints.
- Mechanistic integration strictly eliminates physically impossible negative life predictions.

- Synergistic couplings of joint geometry and material properties dominate complex fatigue behavior.
- Physics-guided architecture outperforms purely unconstrained data-driven models by over 50%.

Abstract

Predicting the fatigue life of welded joints represents a formidable challenge, inherently confounded by intricate multi-physics interactions and the systemic limitations of standalone empirical or purely data-driven models. To circumvent these fundamental bottlenecks, this study proposes a novel physics-informed data-driven framework. Founded upon a comprehensively curated multi-scale fatigue dataset, the proposed paradigm tackles the intricacies of multivariable-coupled service environments by pioneering a physics-data dual-driven deep learning architecture. The crux of this innovation lies in the direct embedding of objective physical fatigue laws—formulated as penalty constraints within the loss function—synergized with physics-guided feature engineering. This rigorous mechanistic integration actively restricts the mathematical hypothesis space, compelling the neural network to converge upon physically plausible mappings rather than unconstrained mathematical correlations. Comprehensive validation demonstrates that this optimized methodology achieves exceptional predictive fidelity and robustness, significantly eclipsing conventional machine learning approaches. Ultimately, this framework successfully harmonizes empirical damage mechanics with advanced artificial intelligence, yielding a highly reliable and interpretable tool for the structural integrity assessment of critical engineering welded components.



Keywords

Fatigue life prediction; Welded joints; Data-driven approach; Physics-informed model; Hybrid algorithm

1. Introduction

The relentless advancement of industrial technology continuously elevates the performance requirements for major engineering equipment and critical structures [1], [2]. In this context, structural reliability is paramount, with fatigue fracture standing as one of the primary failure modes for metal components and systems [3], [4]. Welding, as an indispensable manufacturing technology for engineering structures, often introduces regions of stress concentration at the joints due to geometric and metallurgical discontinuities [5], [6], [7]. Consequently, the fatigue performance of welded joints is typically inferior to that of the base material, making their reliability a critical determinant of the overall structural integrity and service life [8].

The welding process itself is a highly nonlinear, multi-physics coupling phenomenon, and fatigue crack initiation and propagation are influenced by a complex interplay of factors, including material properties, joint geometry, loading conditions, and service environment [9], [10], [11]. Despite decades of research, accurate fatigue life prediction for welded joints remains highly challenging [12]. The fatigue behavior is governed by intricate multi-physics interactions and multi-factor coupling effects under complex operational conditions.

Although conventional empirical models and numerical methodologies remain ubiquitous in fatigue assessment, their efficacy is frequently circumscribed by prohibitive experimental demands and restricted operational applicability. To render complex multiphysics problems tractable, these traditional mechanical approaches inevitably rely on idealized simplifying assumptions. Such simplifications not only compromise prognostic fidelity but also fail to encapsulate the authentic fatigue evolution under multivariable coupling conditions, ultimately culminating in suboptimal generalizability. Conversely, while contemporary purely data-driven architectures demonstrate formidable prowess in nonlinear feature extraction, their intrinsic “black-box” epistemology fundamentally precludes physical interpretability. This mechanistic opacity introduces profound systemic vulnerabilities when models are extrapolated to uncharted operational regimes, thereby impeding their deployment in safety-critical engineering contexts.

Consequently, a compelling impetus exists for a paradigm shift: a synergistic framework capable of leveraging the computational agility of deep learning whilst being rigorously constrained by objective physical mechanics. To address these fundamental

challenges, this study innovatively proposes a physics-data dual-driven framework tailored for the fatigue life prediction of engineering welded structures. Initially, a comprehensive multi-scale fatigue dataset is established to construct baseline hybrid intelligent models. More critically, to tackle intricate multi-factor coupling scenarios, this research pioneers a novel Deep Convolutional Neural Network (DCNN) architecture that deeply integrates welding mechanics through physics-guided feature engineering. The primary novelty of this work lies in creatively embedding classical physical fatigue laws directly into the network's loss function as penalty terms. This deep mechanistic integration forces the algorithm to strictly adhere to actual physical evolution laws during the optimization process. By fundamentally shattering the “black-box” barrier, this methodology endows the data-driven engine with explicit physical interpretability, ensuring that all predictions are not only highly accurate and robust but also strictly physically plausible for complex structural integrity assessments.

2. Related works

This section systematically reviews the state-of-the-art in fatigue life prediction methodologies for welded structures, broadly categorizing them into traditional physics-based models and emerging data-driven machine learning approaches. A critical analysis of the inherent limitations within these existing paradigms is subsequently presented to highlight the fundamental research gap addressed by this study.

2.1. Traditional physics-driven models: mechanistic but rigid

Traditional fatigue assessment primarily relies on deterministic physical or empirical models, including the nominal stress approach [13], hot spot stress method, and critical plane theories (e.g., Morrow [14], Smith Watson Topper (SWT) model [15], Ince-Glinka (IG) models[16], [17] and Chu [18]). The primary strength of these models lies in their rigorous theoretical foundation and clear physical interpretability. They successfully account for fundamental mechanisms such as mean stress effects and local plastic damage.

However, their practical application in complex engineering scenarios is constrained by significant bottlenecks. Welded joints are characterized by multi-physics coupling, including high non-linearity in material gradients, residual stress redistribution, and geometric discontinuities. Traditional models are often too idealized, struggling to characterize the synergistic interactions of these multi-scale factors. Consequently, when applied to real-world components that deviate from standardized laboratory conditions, these rigid formulations frequently exhibit suboptimal generalization and significant prediction dispersion.

2.2. Standalone data-driven models: powerful but “physically blind”

To navigate the complexities introduced by multi-physics coupling, the emergence of data-driven methods has provided a powerful new paradigm for the field of fatigue prediction [3], [19], [20]. In recent years, machine learning algorithms—including Artificial Neural Networks (ANN) [21], [22], [23], Genetic Algorithm – Back Propagation Neural Network (GA-BPNN) [24], [25], Support Vector Machines (SVM) [26], [27], Random Forests (RF) [28], [29], and Convolutional Neural Networks (CNN) [30], [31], along with their hybrid variants—have demonstrated immense potential in characterizing the intricate fatigue behavior of welded joints. Unlike traditional empirical models, these data-driven approaches autonomously learn directly from massive volumes of experimental and operational data. This capability allows them to successfully mine and establish complex non-linear mapping patterns that are exceedingly difficult to describe accurately using traditional explicit mathematical formulations. Despite possessing robust non-linear mapping capabilities, standalone data-driven methods encounter numerous challenges in practical applications, including susceptibility to overfitting, poor convergence, and restricted model interpretability [32], [33].

Nevertheless, the “black-box” nature of standalone machine learning presents a major hurdle for engineering deployment. These models are “physically blind”—they prioritize mathematical correlation over causal mechanical laws. In the absence of physical constraints, data-driven networks are highly susceptible to overfitting localized noise and may generate physically impossible outcomes (e.g., negative life values or violations of damage accumulation boundaries), especially in data-sparse regions. This lack of interpretability and “mechanical soul” severely undermines the confidence required for the safety-critical assessment of major engineering structures.

2.3. Research gap and the proposed integration

The fusion of knowledge-driven mechanisms and data-driven algorithms has recently demonstrated transformative potential across various scientific and engineering disciplines. Knowledge-driven methods excel in theoretical interpretability and structural stability by building causal relationships based on physical mechanisms, whereas data-driven models are adept at capturing complex nonlinear mappings but often lack physical meaning. Integrating the two leverages both the global structural constraints of physics and the local feature extraction prowess of machine learning. This synergistic approach has been successfully implemented in multiple fields. For instance, in medical diagnostics, researchers have utilized domain knowledge to guide the attention mechanisms and feature extraction of deep neural networks (e.g., in CT or MRI image analysis [34]), effectively preventing overfitting under limited datasets while reducing computational complexity. Similarly, in electrical and control engineering [35], studies have successfully embedded explicit thermodynamic laws or implicit system knowledge as

regularization terms within the loss functions of neural networks, thereby guiding policy learning and significantly enhancing both learning efficiency and system performance.

In the realm of materials science and structural integrity, preliminary explorations into dual-driven paradigms have also emerged. For example, in predicting the burst pressure of corroded pipelines, researchers have successfully derived novel features based on mechanical models—incorporating geometric shapes, defect dimensions, and material properties—which effectively elevated the accuracy of the prediction models [36]. However, despite these advances, a synthesis of the current landscape reveals a critical “Missing Bridge”: the lack of a framework that deeply harmonizes the rigorous interpretability of physical laws with the immense computational flexibility of deep learning. Existing research is often polarized, choosing either the rigid precision of physics or the unconstrained flexibility of data. In structural assessment, existing integrations frequently remain at a rudimentary, shallow-coupling stage—typically manifesting as a “series mode” where physical knowledge is solely used for initial feature selection, followed by a standard data-driven prediction. There is a distinct lack of deep, bidirectional synergy.

Furthermore, directly transplanting these existing shallow fusion frameworks into the fatigue life prediction of welded structures is fundamentally inadequate. Welded joints exhibit highly complex nonlinear fatigue behaviors characterized by exceptional data dispersion and the inherent stochasticity of welding processes. More importantly, there are severe disparities between controlled, idealized laboratory testing environments and the multivariable-coupled, complex operational conditions actual structures face in the field.

The primary research contribution of this study is to break this binary limitation and transcend conventional shallow integration modes. By introducing a novel, highly applicable dual-driven paradigm specifically engineered for the fatigue life prediction of engineering welded structures, this work implements a physics-informed integration strategy that embeds objective fatigue laws directly into the neural network's optimization architecture. This multi-level intervention ensures that the data-driven “engine” is strictly regularized by a “mechanical governor,” enabling the model to achieve not only high-precision fitting but also rigorous adherence to the underlying principles of welding mechanics. Consequently, this dual-driven approach effectively addresses the stochasticity of welding processes while maintaining robust physical consistency across complex, unseen operational conditions.

3. Research methodology and data foundation

This section delineates the comprehensive research methodology and the empirical data foundation formulated to address the intricate multi-physics fatigue behavior of engineering welded structures. It sequentially articulates the overarching research objectives, the multi-phase technical approach, and the rigorous compilation of a multi-scale fatigue performance dataset. By systematically integrating multi-dimensional parameters—encompassing varying welding methods, joint geometries, and loading conditions—this section establishes a robust empirical baseline. Furthermore, it charts a clear developmental trajectory from the initial baseline hybrid modeling to the advanced physics-data dual-driven architecture, ultimately serving as the foundational blueprint for the subsequent computational and predictive analyses.

3.1. Research objective

The fundamental novelty of this study lies in pioneering a tailored physics-data dual-driven paradigm specifically for the fatigue assessment of engineering welded structures, effectively bridging the critical gap where generic cross-domain methodologies fall short. While dual-driven concepts have emerged in other scientific fields, their implementation within the intricate fatigue mechanisms of welded joints remains significantly underdeveloped—particularly regarding the integration of domain-specific physical constraints.

This research transcends mere methodological transplantation by explicitly engineering a framework that accounts for the severe disparities between idealized laboratory datasets and the multivariable-coupled complexity of real-world operational environments. Crucially, the proposed model acknowledges the high dispersion and stochasticity inherent in fatigue behavior, translating objective physical laws—such as non-negative life reality and mechanistic error bounds—into dynamic penalty terms directly within the deep learning loss function. This novel architecture ensures the data-driven “engine” is strictly regularized by a “mechanical governor” rooted in welding mechanics. By forcing the neural network to learn strictly physically plausible mappings, this approach not only prevents physically implausible predictions in data-sparse regions but also achieves significant improvements in predictive accuracy and robustness under complex, previously unseen operational conditions.

3.2. Research approach

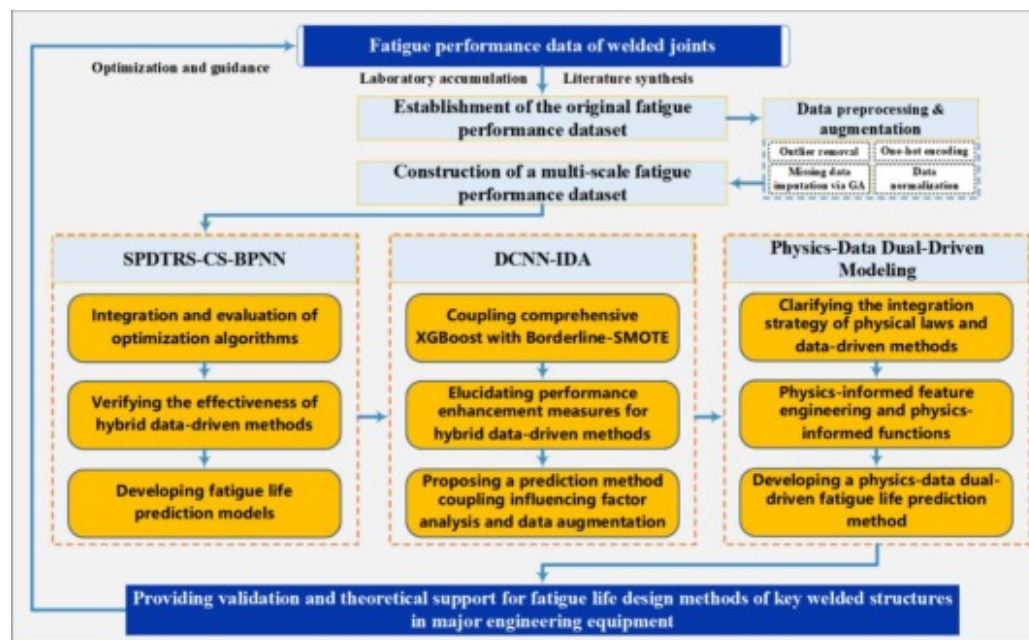
To achieve the overarching objective, the research methodology is systematically structured into three interconnected phases:

- (1) **Multi-Scale Fatigue Dataset Construction.** A comprehensive data foundation is established by integrating extensive multi-scale fatigue test data of welded joints. This phase involves rigorous data processing to

capture the complex, multi-factor coupling conditions inherently present in actual engineering operations.

- (2) **Baseline Hybrid Intelligent Modeling.** Under standard operational conditions, a baseline hybrid machine learning model is formulated. This step serves to capture the fundamental non-linear relationships between structural features and fatigue life, verifying the feasibility of the data-driven mapping.
- (3) **Physics-Data Dual-Driven Model Development.** To address scenarios involving highly complex multi-factor coupling, a DCNN is developed. In this critical phase, physical fatigue mechanisms are mathematically translated into penalty terms within the loss function during the backpropagation training. This physically-guided optimization restrains the network's weight updates, ensuring that all subsequent fatigue life predictions remain physically plausible and strictly adhere to underlying fatigue laws.

A comprehensive visual summary of the proposed research framework and the corresponding methodology is illustrated in Fig. 1.



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Fig. 1. Flowchart summarizing the proposed physics-informed data-driven fatigue life prediction framework.

3.3. Construction of multi-scale fatigue performance dataset

The predictive fidelity and generalization capability of any data-driven framework are intrinsically governed by the quality, diversity, and comprehensiveness of its foundational dataset. To effectively capture the intricate multi-physics coupling effects and multivariable interactions inherent in engineering welded joints, this section delineates the systematic construction of a multi-scale fatigue performance dataset.

3.3.1. Construction of the original dataset

The generalizability of a data-driven approach for predicting the fatigue life of welded joints is a critical metric for assessing its performance. To ensure broad applicability, the proposed method must accommodate diverse operational scenarios. This study integrates multi-scale factors—including material type, geometry, service environment, loading conditions, joint configuration, and welding parameters—into a unified fatigue performance dataset. This comprehensive approach enables a more robust understanding and prediction of the fatigue life of welded joints under varying conditions.

To guarantee data quality, the data collection adhered to two key principles:

- (1) The dataset must adequately represent the value ranges of all influencing factors.
- (2) The fatigue performance data must be consistent, complete, and reliable across different fatigue conditions.

By consolidating fatigue test data from in-house laboratory experiments and data from published literature [37], [38], [39], [40], [41], [42], [43], an original fatigue performance dataset was established. The final dataset comprises 189 S-N curves, encompassing a total of 2,984 individual fatigue data points.

3.3.2. Data preprocessing

- (1) Outlier Removal.

The removal of outliers that contradict physical principles is essential for aligning the model's training and predictions with actual fatigue behavior. This process not only simplifies model training but also enhances the model's generalizability to new, unseen data. For this purpose, Eq. (1), (2) were established as criteria to screen the original dataset. Any data point failing to satisfy either of these two equations was deemed an outlier and removed.

Specifically, Eq. (1) was applied to eliminate data points where the deviation of the fatigue life value from the S-N curve exceeded five times the average deviation of all data points in the dataset. Eq. (2) was used to remove data points where the deviation of the fatigue life value from the S-N curve value exceeded one order of magnitude (a factor of 10).

$$|N_{\text{point},c} - N_{S-N,c}| < \frac{5}{n_a} \sum_{i=1}^{n_a} (|N_{\text{point},i} - N_{S-N,i}|) \quad (1)$$

$$\frac{1}{10} < \frac{N_{\text{point},c}}{N_{S-N,c}} < 10 \quad (2)$$

here, $N_{\text{point},c}$ is the fatigue life value under scrutiny, $N_{S-N,c}$ is the fatigue life value on the S-N curve corresponding to the same stress range as the data point in question, and n_a is the total number of fatigue performance data points in the original dataset. $N_{\text{point},i}$ and $N_{S-N,i}$ represent the i fatigue life data point and the corresponding fatigue life value on the S-N curve at the same stress range, respectively.

Following this procedure, 252 data points that were inconsistent with the physical principles of fatigue were removed. This outlier removal process enhances the stability of the fatigue life prediction model during training and improves the accuracy and reliability of the model's predictions.

(2) Numerical Encoding of Categorical Features.

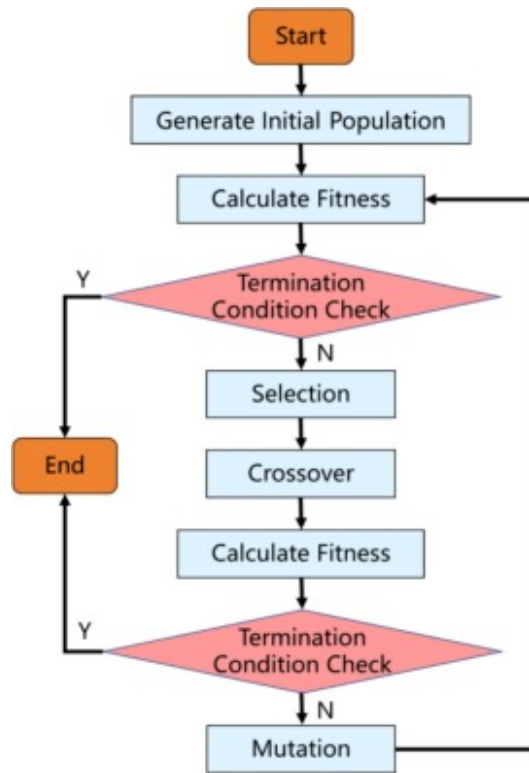
Data-driven algorithms require all input features to be in numerical form. Consequently, discrete categorical features, such as welding method, material type, and loading mode, must be converted into numerical representations [44]. This study employs the one-hot encoding technique for this conversion. This method maps discrete categorical features into a higher-dimensional space, effectively transforming each category into a binary vector of length n , where only one element is 1 (representing the active category) and all others are 0. This approach prevents the model from incorrectly interpreting an inherent order or magnitude among the categories, which could lead to erroneous analysis. The categorical features processed using this method, along with their respective possible values, are summarized in Table 1.

Table 1. Selection range of category features contained in the dataset.

Category characteristics	Selection range
Joint type	Butt joint; T-joint; Cross joint
Welding method	Flux Core Arc Welding (FCAW); Sub Arc Welding (SAW); Gas Metal Arc Welding (GMAW); Plasma Arc Welding (PAW); Shielded Metal Arc Welding (SMAW); Gas Tungsten Arc Welding (GTAW); Laser Beam Welding (LBW)
Material type	Steel (EH36, Q235, Q420, 316H); Titanium alloy (TA1, TA2, TC4, TC6); Aluminum alloy (2219, 5052, 6061, 7075); Magnesium alloy (AZ31B, AZ91D, ZK61M)
Loading form	Tension-compression load; Bending load; Torque load
Surface condition	No processing; Coating; Shot peening; Nitriding

(3) Missing Data Imputation

A portion of the fatigue dataset compiled from published literature contained missing values for certain influence factors. Such omissions can impede model convergence and undermine the reliability of subsequent factor analyses and life predictions [45], [46]. To address this, a Genetic Algorithm (GA) was employed to impute the missing data. As a heuristic search method, the GA is highly versatile and problem-agnostic, making it exceptionally well-suited for this task. The specific imputation workflow is illustrated in Fig. 2.



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Fig. 2. Schematic diagram of genetic algorithm.

(4) Data Normalization

The various fatigue performance features exhibit significant differences in their value ranges. For instance, the stress range typically falls within $(0, 1 \times 10^3)$ MPa, while the fatigue life generally lies within $(1 \times 10^3, 1 \times 10^7)$ cycles. This substantial disparity in numerical scales makes it difficult to compare their relative variations and can adversely affect model training. Therefore, normalizing the dataset is necessary. The Z-score normalization method [47], [48] addresses this by scaling each feature based on the dataset's mean and standard deviation, thereby mapping the data to a comparable, dimensionless scale. The normalized

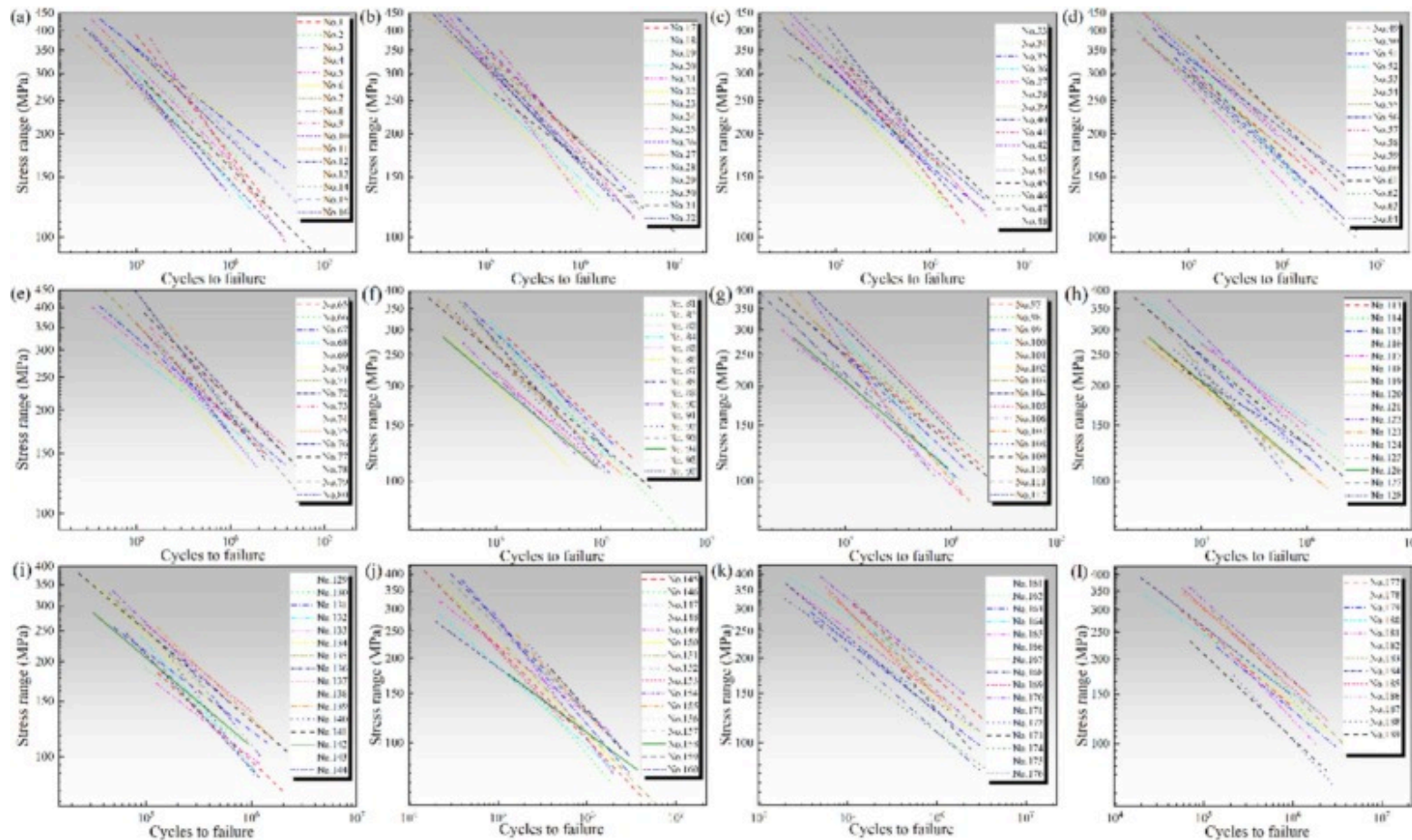
value for each fatigue performance feature is calculated as follows:

$$Value' = \frac{Value - Ave_d}{Std v_d} \quad (3)$$

where Value' is the normalized feature value, Value is the original feature value, Ave_d is the mean of all values for that feature in the dataset, and Std_{v_d} is the corresponding standard deviation.

3.3.3. Dataset statistics

Following the procedures of screening, cleaning, and preprocessing, the established multi-scale fatigue performance dataset comprises 189 S-N curves, corresponding to a total of 2,732 individual data points. The dataset was partitioned into a training set (80%) and a test set (20%) to ensure model generalizability and prediction accuracy. The dataset contains a substantial number of fatigue data points. The S-N curves, derived based on the nominal stress approach, provide a clear representation of the relationship between stress levels and fatigue life, offering key insights into the fatigue behavior of welded joints, as illustrated in [Fig. 3](#). The distribution of various categorical features within the dataset, as summarized in [Table 1](#), underscores its diversity and representativeness, thereby providing a robust foundation for reliable fatigue life prediction.



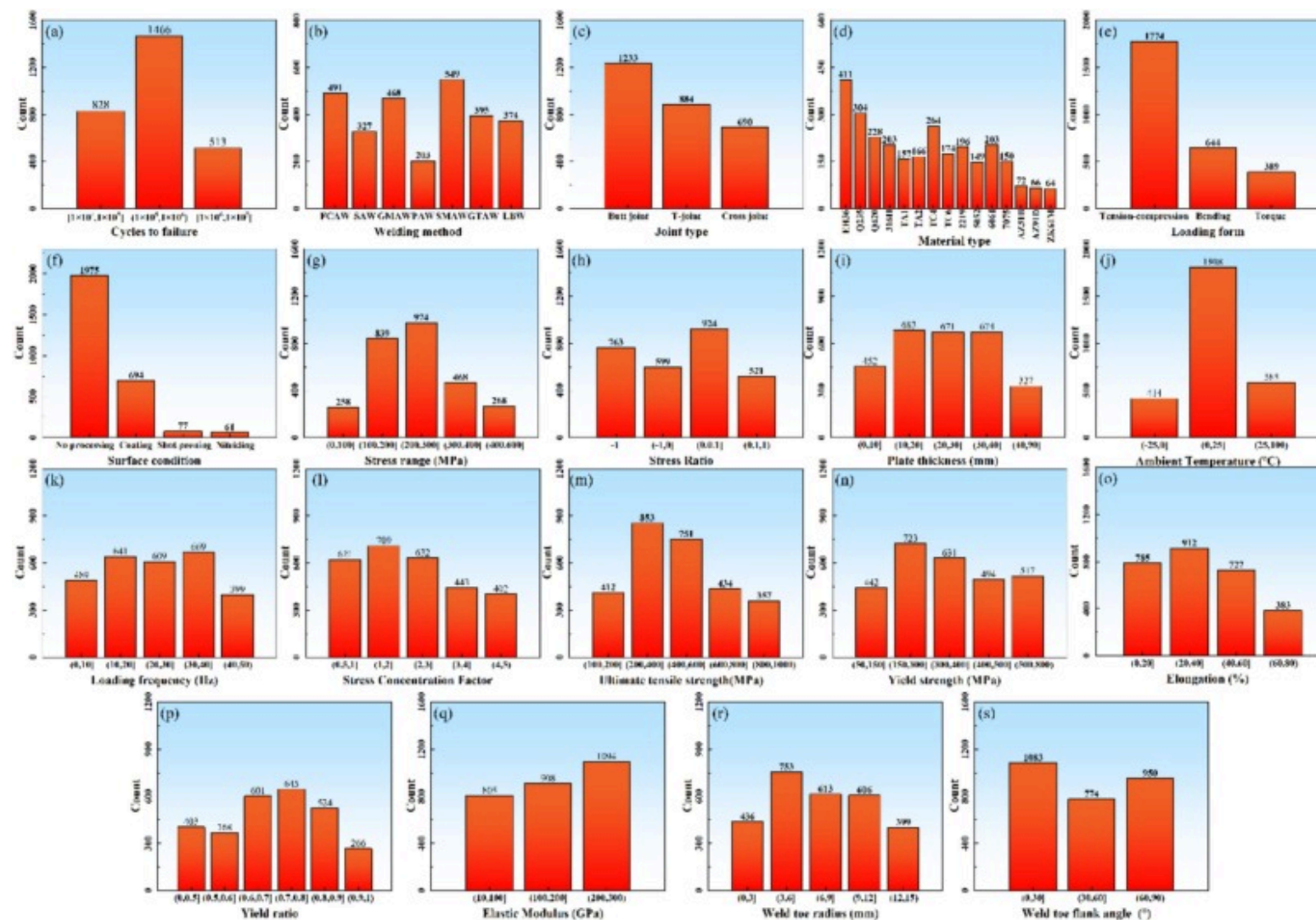
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Fig. 3. Set of S-N curves obtained based on nominal stress method: (a–e) Sample Nos. 1–80 with butt joints, (f–i) Sample Nos. 81–144 with T-type joints, and (j–l) Sample Nos. 145–189 with cruciform joints.

The ranges, specific distributions, and corresponding characteristic information for the different fatigue performance features in the established multi-scale dataset are presented in Fig. 4. Table 2 summarizes the distribution, mean, and standard deviation of these features. Taking fatigue life as an example, 52.2% of the data points fall within the range of $(1 \times 10^5, 1 \times 10^6)$ cycles, with a mean value of 9.7×10^5 cycles and a standard deviation of 1.26×10^6 . The considerable variability in fatigue life, indicated by the

high standard deviation, arises from differences in materials, manufacturing processes, and testing conditions across the source literature. This reflects the inherent complexity of welded joint fatigue performance and underscores the necessity of incorporating multi-scale factors into the predictive model.



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Fig. 4. Distribution of the range of values for various influencing factors of fatigue life: (a-s) are fatigue life, welding method, joint type, material type, loading form, surface condition, stress range, stress ratio, plate thickness, temperature, frequency, stress concentration factor, tensile strength, yield strength, elongation, yield ratio, elastic modulus, weld toe radius and weld toe flank angle.

Table 2. Influencing factors of fatigue life contained in the fatigue performance dataset.

Factors affecting fatigue life	Range of values	Average value	Standard deviation
Stress Range, $\Delta\sigma$ (MPa)	(0, 600)	257.4	91.4
Stress Ratio, R	[-1, 1)	-0.2	0.7
Plate Thickness, t (mm)	(0, 90]	26.7	22.6
Ambient Temperature, T (°C)	(-25, 100)	26.8	32.9
Loading Frequency, f (Hz)	(0, 50)	19.8	17.7
Stress Concentration Factor, (Kt)	(0.5, 5)	2.1	1.9
Ultimate Tensile Strength, (MPa)	(100, 1000)	488.3	230.6
Yield Strength, YS (MPa)	(50, 800)	271.4	91.3
Elongation, δ (%)	(0, 80)	19.1	13.7
Yield Ratio, YTR	(0, 1)	0.6	0.3
Elastic Modulus, E (GPa)	(10, 300)	168.7	60.1
Weld Toe Radius, r (mm)	(0, 15)	6.7	4.9
Weld Toe Flank Angle, θ (°)	(0, 90)	40.3	26.7

Regarding the categorical features, the SMAW process constitutes the largest portion of the dataset (19.6%), while PAW is the least represented (7.2%). GTAW and GMAW account for 16.7% and 14.1%, respectively. Butt joints represent 43.9% of the data, and

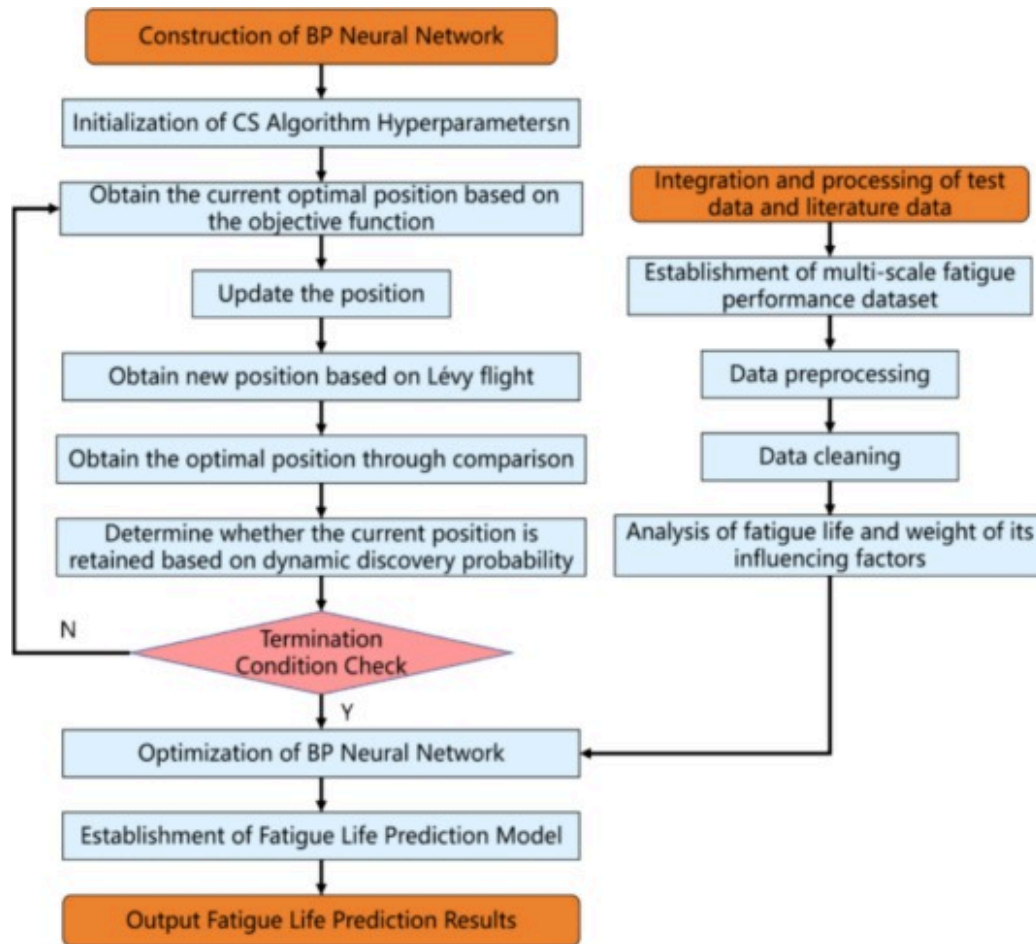
tests conducted in an air environment account for 49.3%. In terms of materials, joints of steel, titanium alloy, aluminum alloy, and magnesium alloy constitute 40.8%, 27.1%, 24.9%, and 7.2%, respectively. Load-controlled testing under tension–compression conditions represents 63.2% of the data. Data for specimens without surface treatment and without coating account for 70.4% and 24.7%, respectively. The broad distribution and detailed statistics of the dataset demonstrate its diversity and comprehensiveness, thereby providing a solid foundation for reliable fatigue life prediction.

4. Fatigue life prediction model based on the SPDTRS-CS-BPNN hybrid data-driven approach

Predicting the fatigue life of welded joints using a single data-driven method presents significant challenges. Each algorithm possesses inherent limitations and a specific scope of applicability, often struggling to capture the complete spectrum of complex, non-linear relationships involved. Furthermore, solitary models are prone to overfitting the training data, resulting in poor generalizability when applied to new, unseen datasets. To overcome these limitations, this paper proposes a hybrid data-driven framework named SPDTRS-CS-BPNN. This integrated framework synergistically combines a Sensitivity Parameter Determination and Tuning Reduction System (SPDTRS) for feature weighting analysis, a Cuckoo Search (CS) algorithm for parameter optimization, and a Back-Propagation Neural Network (BPNN) as the core predictor. The objective is to achieve accurate and robust predictions of the fatigue life for EH36 high-strength steel welded joints.

4.1. The SPDTRS-CS-BPNN hybrid methodology

The proposed hybrid framework is designed to leverage the complementary strengths of its constituent components. The SPDTRS module first conducts a theoretical analysis to determine the weight of each fatigue life influencing factor. This analysis provides a valuable initialization strategy for the BPNN, effectively weighting its inputs to ensure that the influence of different factors is appropriately prioritized within the prediction model. Concurrently, the CS algorithm performs an automated search to optimize the key architectural parameters of the BPNN, such as the learning rate and the number of neurons in the hidden layer, aiming to identify the most effective model configuration. During this optimization process, the CS algorithm seeks the optimal solution that balances multiple performance metrics, such as prediction accuracy and computational efficiency. The workflow of this integrated methodology is illustrated in [Fig. 5](#), with the key steps detailed as follows:



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Fig. 5. Fatigue life prediction approach via hybrid data-driven algorithms.

Step 1: Identification of Key Influencing Factors and Data Preprocessing.

Based on prior research and analysis of the established multi-scale dataset [49], [50], [51], five key factors were identified for predicting the fatigue life of EH36 steel welded joints: welding method (M_{weld}), plate thickness (T_{plate}), joint type (T_{joint}), stress ratio (R), and stress range ($\Delta\sigma$). The corresponding EH36 steel data subset was extracted from the multi-scale dataset and

subjected to data cleaning. The input–output mapping for the established fatigue life prediction model is defined as: $[M_{\text{weld}}, T_{\text{plate}}, T_{\text{joint}}, R, \Delta\sigma] \rightarrow N_p$ where N_p is the predicted fatigue life.

Discrete categorical variables were converted into numerical features using one-hot encoding to prevent the model from incorrectly assuming an ordinal relationship. For instance, the T_{joint} possesses three possible categories (x_{T1} , x_{T2} , x_{T3}). After one-hot encoding, it is represented by a vector V_T of length 3. For any given category, the element at the corresponding index i in V_T is set to 1, while all other elements are 0. The one-hot encoding for T_{joint} can therefore be expressed by the following formulation:

$$\vec{V}_T = \{v_{T1}, v_{T2}, v_{T3}\} \quad (4)$$

where $v_{Ti} = 1$ if $T_{\text{joint}} = x_{Ti}$, and $v_{Ti} = 0$ otherwise.

Step 2: Weight Analysis of Influencing Factors based on SPDTRS.

The SPDTRS is employed to analyze the interrelationships and quantify the relative weights of the influencing factors on fatigue life. This quantitative analysis provides critical guidance for the subsequent training and optimization of the BPNN. Specifically, a data-driven matrix, constructed from the loss functions defined in Eqs. (5) to (10), facilitates a comprehensive synthesis of the weights associated with different factors. This process elucidates the underlying influence mechanisms of various factors on fatigue life, thereby enabling the prediction model to better comprehend the complex, multifactorial, and non-linear relationships between fatigue life and its influencing parameters.

$$\lambda_{PP} = 0 \quad (5)$$

$$\lambda_{BP} = S(X|[x])(P(X|[x]) - \zeta) \quad (6)$$

$$\lambda_{NP} = S(X|[x]) \quad (7)$$

$$\lambda_{PN} = S^C(X|[x]) \quad (8)$$

$$\lambda_{BN} = S^C(X|[x])(1 - P(X|[x]) - \zeta) \quad (9)$$

$$\lambda_{NN} = 0 \quad (10)$$

Step 3: BPNN Parameter Optimization using the CS Algorithm.

The weights derived from the SPDTRS analysis were used to initialize the connections of the BPNN's input layer. The BPNN architecture was configured with an input layer of five neurons (corresponding to M_{weld} , T_{plate} , T_{joint} , R and $\Delta\sigma$), a single output neuron (representing the predicted fatigue life N_p), and a hidden layer whose size was preliminarily determined through experimentation to contain eight neurons. The hyperparameters of the CS algorithm were initialized. The CS algorithm was then employed to enhance the BPNN's parameter optimization and solution-seeking capabilities, utilizing an objective function to identify the optimal set of BPNN parameters under the current conditions.

Step 4: Model Training and Output.

The current best solution obtained in Step 3 was used to update the candidate solutions' positions. New positions were generated via Lévy flights to facilitate global exploration and local refinement. A dynamic discovery probability was applied to either retain the best solutions or replace inferior ones. By evaluating the fitness of these new positions, a new optimal solution was identified. This solution was subjected to further analysis based on the dynamic discovery probability to determine its retention. If the termination criteria (e.g., a maximum number of iterations or a minimum error threshold) were not met, the process reverted to generating new positions via Lévy flights; otherwise, the optimal solution was output.

Step 5: Establishment of the SPDTRS-CS-BPNN Fatigue Life Prediction Model.

The final model was established based on the hybrid SPDTRS-CS-BPNN data-driven algorithm. Specifically, the partial derivatives of the error function with respect to the neurons in the hidden layer, $\delta_h(k)$, and the output layer, $\delta_o(k)$, were calculated based on the difference between the BPNN's expected and actual outputs. These derivatives, along with the inputs to the hidden and output layer neurons, were then used to adjust and optimize the connection weights between the input and hidden layers (w_{ih}) and between the hidden and output layers (w_{ho}). The global error E_t for the epoch was calculated according to the following equation:

$$E_t = \frac{1}{2m} \sum_{k=1}^m \sum_{o=1}^q (do(k) - yo(k))^2 \quad (11)$$

The iterative process of the BPNN is terminated when the global error E_t meets the predefined computational accuracy ϵ or when the maximum number of learning iterations M is reached. Upon satisfying either criterion, the development of the welded

joint fatigue life prediction model is considered complete. The finalized model can then reliably predict the fatigue life of EH36 steel welded joints by receiving the corresponding values of the input influencing factors.

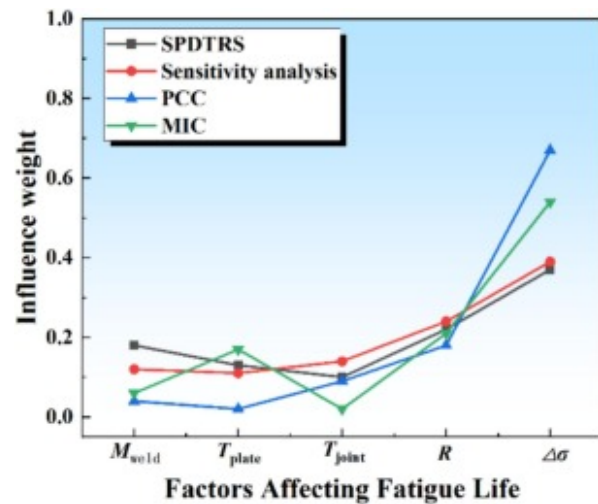
4.2. Weight analysis of fatigue life influencing factors

To robustly validate the performance and generalization capability of the established model, it was evaluated against a strictly independent set of parameter combinations completely excluded from the training and testing phases. This rigorous verification ensures the model's predictive reliability when exposed to unseen data. The eight distinct parameter configurations (No. 1–No. 8) detailed in Table 3 were strategically selected to reflect the inherent diversity and complexity of welded joints in practical engineering applications. Specifically, these configurations encompass varying combinations of welding methods, plate thicknesses, joint types, and stress ratios. By benchmarking the model's predicted fatigue life against actual experimental values under these independent conditions, its extrapolation capacity and predictive fidelity for novel data can be comprehensively assessed.

Table 3. Samples for model validation.

Number	Welding method	Plate thickness	Joint type	Stress Ratio
No.1	FCAW	32	Butt joint	0
No.2	SMAW	12	Cross joint	−1
No.3	FCAW	24	T-joint	0.1
No.4	SMAW	8	Butt joint	0.1
No.5	GMAW	16	T-joint	0
No.6	GMAW	28	Butt joint	−1
No.7	FCAW	18	T-joint	0.5
No.8	SMAW	12	Cross joint	0.1

To validate the effectiveness of the proposed algorithm, its performance was compared against four established methods: The SPDTRS algorithm, Sensitivity Analysis, Pearson Correlation Coefficient (PCC), and Maximal Information Coefficient (MIC). These methods were employed collectively to analyze the influence weights of the five key factors (M_{weld} , T_{plate} , T_{joint} , R and $\Delta\sigma$). The results are presented in Fig. 6.



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Fig. 6. Analysis of factors influencing fatigue life via data-driven methods.

All methods consistently identified the stress range ($\Delta\sigma$) as the most significant factor influencing the fatigue life of EH36 steel welded joints. The SPDTRS algorithm assigned it a weight of 0.37, while the results from Sensitivity Analysis, PCC, and MIC were 0.39, 0.67, and 0.54, respectively. This consensus underscores the decisive role of load level in fatigue damage accumulation. The stress ratio (R) was identified as the second most influential factor, with its weight (SPDTRS: 0.22) and correlation metrics being substantially higher than those of the remaining factors. This finding aligns with the physical mechanism whereby the stress ratio alters fatigue crack initiation and propagation behavior by influencing the mean stress. In contrast, the effects of the M_{weld} , T_{plate} , and T_{joint} were relatively weaker, yet non-negligible. Collectively, these factors reflect the influence of manufacturing processes and geometric configurations on local stress concentration and potential defects.

A comparison between the findings of this study and existing literature is provided in Table 4. The studies by Yang et al. [52], [53] and Zhang et al. [54] similarly emphasized the primary importance of the stress range and the secondary importance of the stress ratio, which corroborates the conclusions drawn here. However, the proposed SPDTRS-CS-BPNN framework offers a more comprehensive perspective by simultaneously quantifying the effects of a wider variety of factors.

Table 4. Analysis of factors influencing fatigue life.

Sample number	M_{weld}	T_{plate}	T_{joint}	R	$\Delta\sigma$
SPDTRS	0.18	0.13	0.1	0.22	0.37
Sensitivity analysis	0.14	0.11	0.14	0.24	0.39
PCC	0.04	0.02	0.09	0.18	0.67
MIC	0.06	0.17	0.02	0.21	0.54
Yang et al.[52]	0	0	0.33	0	0.67
Yang et al.[53]	0	0	0	0.33	0.67
Zhang et al.[54]	0.17	0	0	0.4	0.43

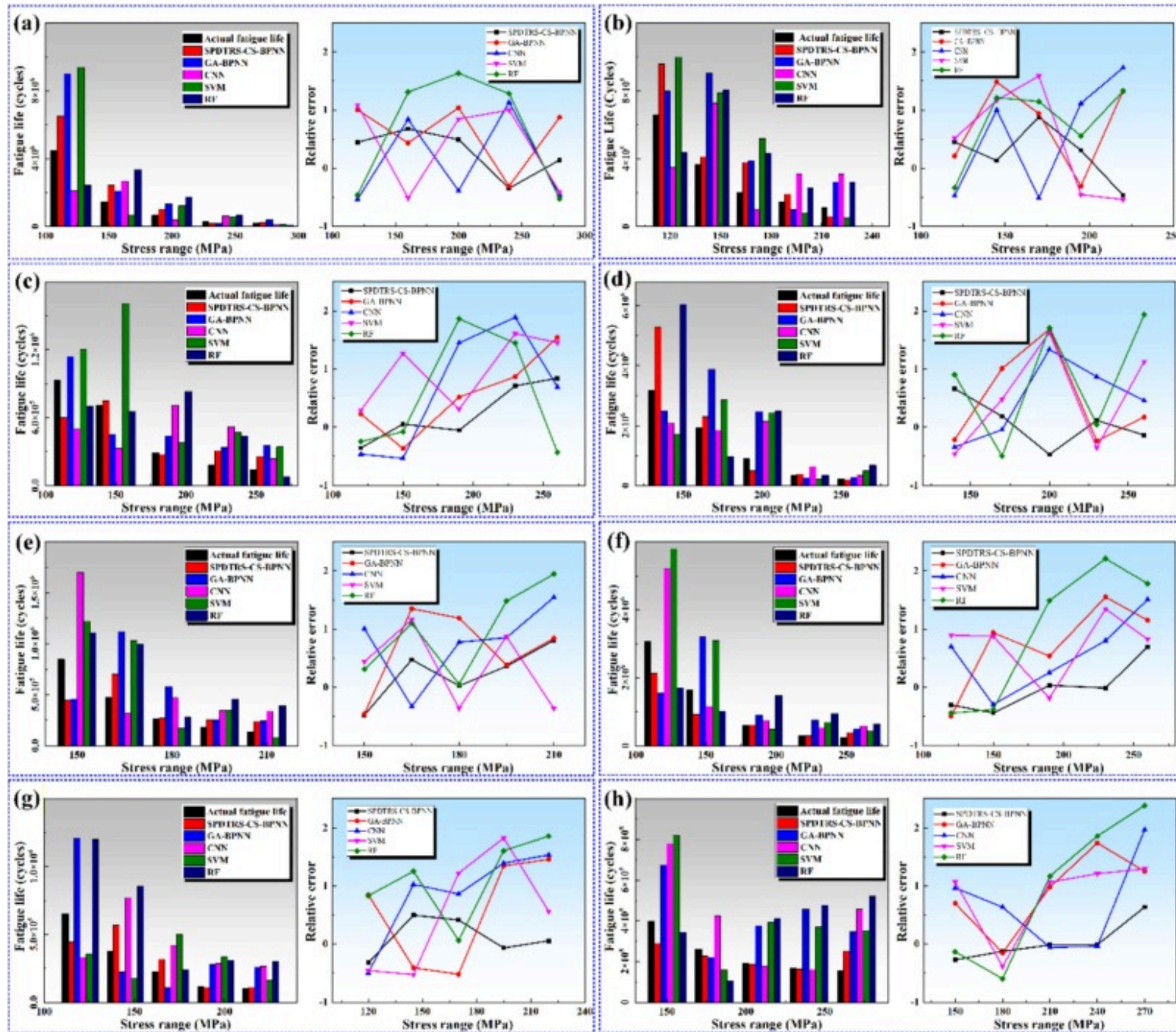
4.3. Model predictive performance and validation

To evaluate the performance of the proposed SPDTRS-CS-BPNN model, a comparative analysis was conducted against four established data-driven models: GA-BPNN, SVM, RF, and CNN. All models were trained and tested using the identical dataset.

4.3.1. Prediction results and error analysis

Fig. 7 presents a comparison between the experimentally measured fatigue lives and the predicted values from different models across eight distinct parameter sets under various stress levels. Visually, the predictions of the SPDTRS-CS-BPNN model demonstrate superior agreement with the experimental data compared to the other models. This suggests that the proposed hybrid framework can more effectively capture the complex effects of various factors influencing welded joint fatigue life. The

integration and complementarity of the constituent algorithms appear to significantly enhance prediction accuracy. In contrast, other models, particularly RF and SVM, exhibit considerable prediction deviations under certain conditions.



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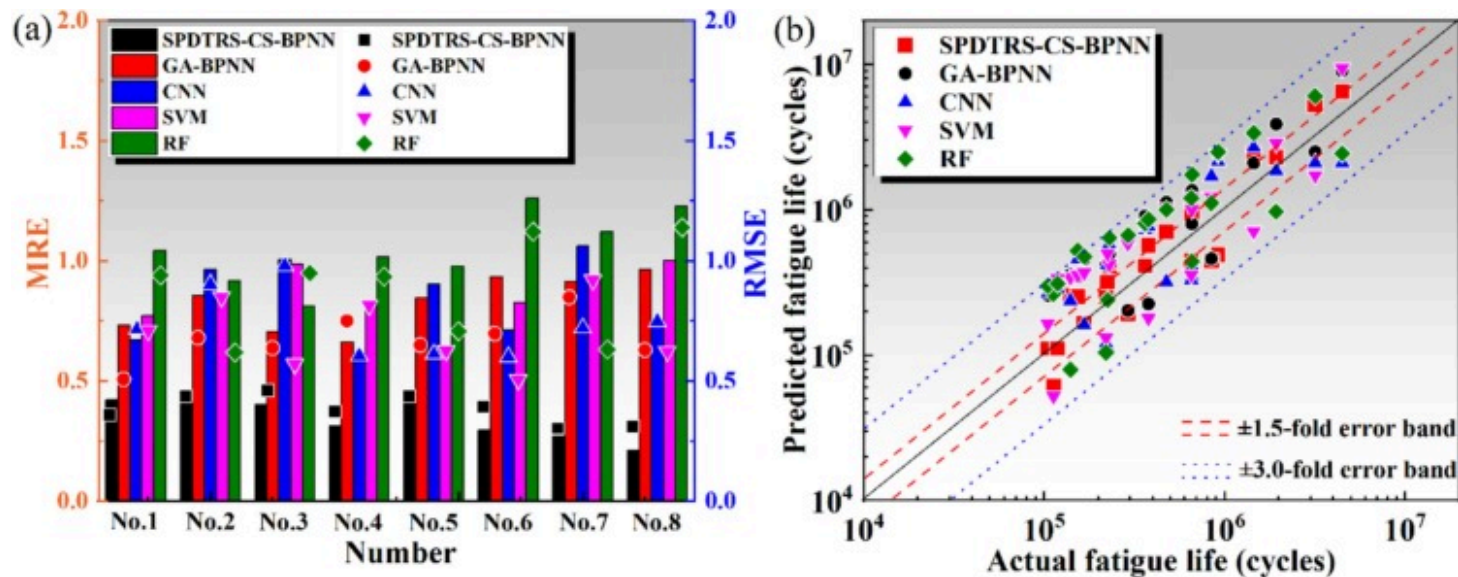
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Fig. 7. Comparison of fatigue life prediction results with actual results and the error thereof: (a) No.1, (b) No.2, (c) No.3, (d) No.4, (e) No.5, (f) No.6, (g) No.7 and (h) No.8.

This visual advantage is further substantiated by quantitative error analysis. As shown in Fig. 7, the maximum relative error (E_{max}) for the SPDTRS-CS-BPNN model remains consistently below 1.0 across all eight test conditions. This performance is notably superior to that of the GA-BPNN model ($E_{max} > 1.5$), the CNN and SVM models ($E_{max} \approx 2.0$), and the RF model ($E_{max} \gg 2.0$).

4.3.2. Comprehensive performance evaluation

Fig. 8 and Table 5 summarize the comprehensive performance metrics for all models. The proposed SPDTRS-CS-BPNN model achieved a maximum relative error (E_{max}) of 0.88, which represents a reduction of 47.6%, 55.3%, 51.6%, and 63.0% compared to the GA-BPNN, CNN, SVM, and RF models, respectively. Furthermore, its Mean Relative Error (MRE) and Root Mean Square Error (RMSE) were 0.35 and 0.38, respectively—the lowest values among all benchmarked models. These results demonstrate its superior predictive accuracy and stability.



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Fig. 8. Performance comparison of different fatigue life prediction methods: (a) prediction error and (b) error band.

Table 5. Comparison of accuracy and stability of different prediction methods.

	SPDTRS-CS-BPNN	GA-BPNN	CNN	SVM	RF
$E_{\max}$	0.88	1.68	1.97	1.82	2.38
MRE	0.35	0.83	0.82	0.85	1.05
RMSE	0.38	0.68	0.74	0.70	0.88

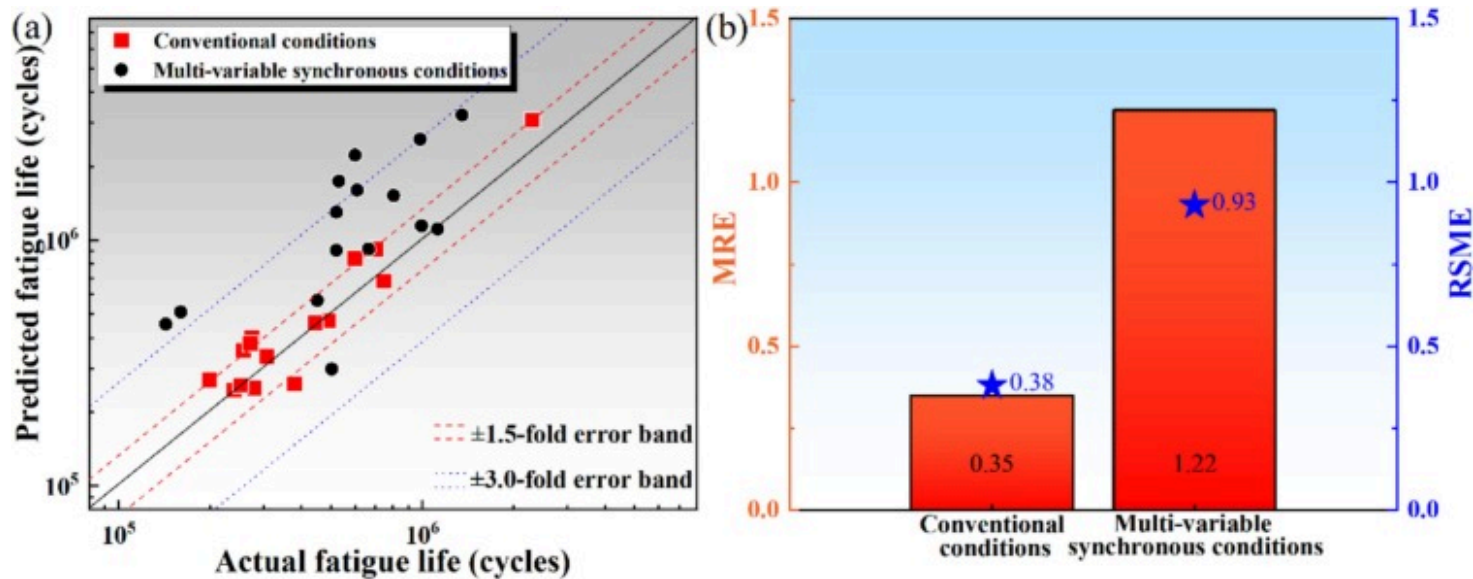
The distribution of prediction errors, illustrated by the error band in Fig. 8b, provides further evidence of the model's robustness. All data points predicted by the SPDTRS-CS-BPNN model fall within the ± 3 error band, with a high concentration near the center and no apparent outliers. In contrast, the predictions from the other models are more widely scattered, with some points even lying outside the ± 3 error band. This contrast confirms that the proposed hybrid model possesses significantly stronger robustness when dealing with the inherent variability and scatter typical of fatigue data.

4.4. Discussion and limitations

The experimental results robustly demonstrate that the SPDTRS-CS-BPNN hybrid model successfully enhances the predictive performance for welded joint fatigue life through the effective integration and complementarity of its constituent algorithms. The SPDTRS component provides a physically meaningful initialization of feature weights, while the CS algorithm prevents the BPNN from converging to local minima. This synergistic framework ultimately enables the BPNN to more accurately learn the complex, non-linear mapping between multiple influencing factors and fatigue life.

Notwithstanding its promising performance, the proposed model possesses certain limitations. A notable decline in predictive capability is observed when the number of influencing factors increases substantially to simulate more realistic, complex marine engineering scenarios characterized by the simultaneous variation of multiple parameters, such as material strength, loading frequency, ambient temperature, and surface condition. As illustrated in Fig. 9, under such highly complex conditions, the

model's MRE and RMSE increased by 248.6% and 144.7%, respectively, with predicted points deviating significantly from the ideal agreement line. This indicates that purely data-driven models still face challenges in generalizing to highly complex operational regimes not sufficiently represented in the training data. Furthermore, the inherent “black-box” nature of data-driven models presents a significant obstacle to their application in safety-critical engineering fields, where model interpretability is paramount. Consequently, future research should be directed toward developing a new generation of fatigue life prediction methodologies that can seamlessly integrate physical priors and offer enhanced interpretability.



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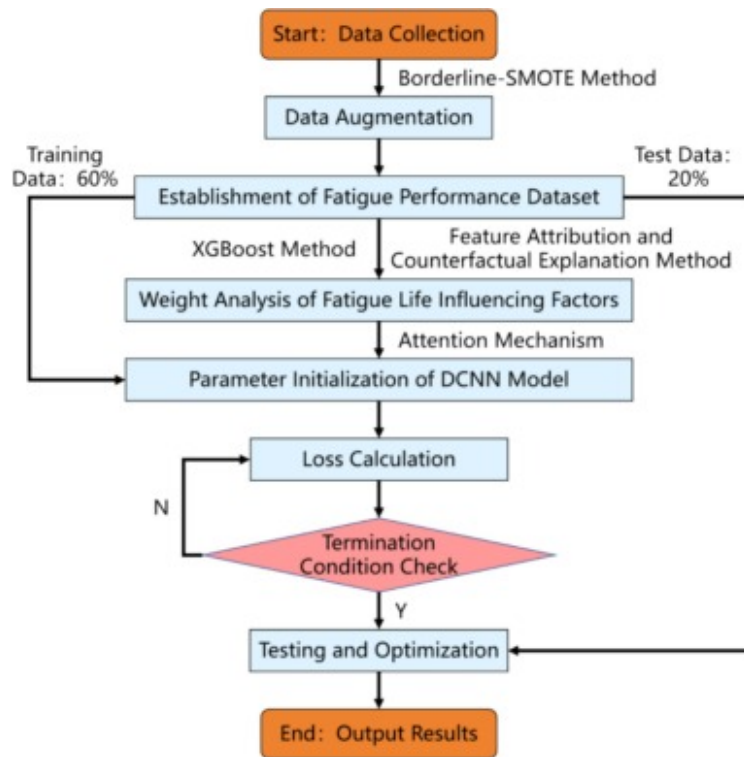
Fig. 9. Limitation analysis of prediction methods: (a) error band and (b) prediction error.

5. Fatigue life prediction using DCNN coupled with influencing factor analysis and data augmentation

To address the performance degradation of the SPDTRS-CS-BPNN model under multi-factor coupling conditions, this section proposes a novel framework that deeply integrates data augmentation and interpretability analysis within a DCNN architecture. The objective of this methodology is to enhance the robustness and reliability of the predictive model in complex service conditions by simultaneously improving dataset quality and model transparency.

5.1. Coupling strategy: integrating factor analysis and data augmentation

The core concept of the proposed method is to establish an iterative optimization loop: Influencing Factor Analysis→Data Augmentation→Model Training. This closed-loop system is designed to continuously refine the model. The technical roadmap is illustrated in Fig. 10, with the specific steps detailed below.



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Fig. 10. Flow chart of hybrid influencing factor analysis-data augmentation approach.

Step 1: Multi-Feature Input Space Expansion.

The input feature space is expanded from the initial five key factors (M_{weld} , T_{plate} , T_{joint} , R and $\Delta\sigma$) to include five additional significant variables: yield strength (σ_s), loading frequency (f), loading mode (F_{loading}), ambient temperature (T_e), and surface

condition (S_{surface}) [13], [19]. Consequently, the input–output mapping for the model is redefined as: $[M_{\text{weld}}, T_{\text{plate}}, T_{\text{joint}}, R, \Delta\sigma, f, F_{\text{loading}}, T_e, S_{\text{surface}}] \rightarrow N_p$. This expansion enables the model to more comprehensively account for the multi-physics factors influencing fatigue life.

Step 2: Data Augmentation via Borderline-SMOTE.

To mitigate the data imbalance problem prevalent in high-dimensional feature spaces, the Borderline-SMOTE algorithm is employed for targeted data augmentation. Unlike simple random interpolation, this algorithm specifically synthesizes new minority-class samples in the decision boundary region. It first identifies “dangerous” minority samples—those whose nearest neighbors are predominantly from the majority class. Then, it generates synthetic samples by interpolating between these identified samples and their nearest neighbors. This strategy effectively augments the data volume in critical, hard-to-classify regions, thereby improving dataset quality and providing a more balanced and representative foundation for DCNN training.

Step 3: Influencing Factor Analysis using XGBoost.

The XGBoost algorithm is utilized for analyzing the weight of influencing factors. XGBoost is a gradient boosting decision tree model whose objective function, as shown in Eq. (12), consists of a loss function and a regularization term, which effectively prevents overfitting:

$$Ob_j^{(k)} \approx \sum_{j=1}^T \left[\left(\sum_{i \in I_j} g_i \right) w^* + \frac{\left(\sum_{i \in I_j} h_i + \lambda \right)}{2} w^{*2} \right] + \gamma T \quad (12)$$

In Eq. (12), the notation $i \in I_j$ indicates that the i -th sample is assigned to the j -th leaf node, where g_i and h_i represent the first-order and second-order partial derivatives (gradient and Hessian) of the loss function, respectively. The parameters λ and γ are regularization hyperparameters that control the penalty strength.

Step 4: Physics-Informed DCNN Model Construction and Training.

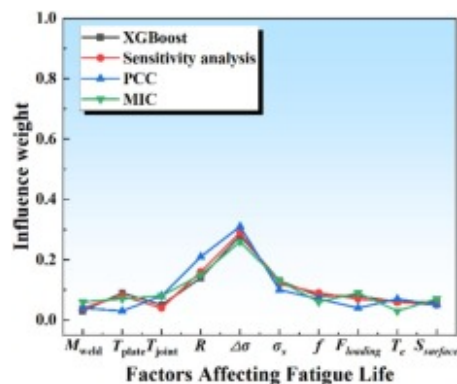
The feature weights derived from the XGBoost analysis are integrated into the DCNN input through an attention mechanism layer, enabling the model to prioritize significant features from the initial training phase. The DCNN architecture comprises multiple convolutional layers, pooling layers, and fully connected layers, leveraging its robust spatial feature extraction capability

to learn complex mappings between high-dimensional features and fatigue life. During training, a dynamic learning rate strategy is employed, and model performance is monitored on a validation set with early stopping to prevent overfitting.

5.2. Analysis of fatigue life influencing factors

To gain deeper insights into the fatigue behavior of welded joints under multi-factor coupling effects, the XGBoost algorithm was employed to assess the importance of the ten key influencing factors.

The importance rankings of the influencing factors derived from the XGBoost algorithm are presented in Fig. 11. The analysis reveals that the stress range ($\Delta\sigma$) remains the most significant factor affecting fatigue life (weight: 0.28), which aligns well with the physical nature of uniaxial fatigue. The stress ratio (R) and material yield strength (σ_s) rank second and third, with weights of 0.14 and 0.13, respectively, highlighting the critical roles of mean stress effects and the inherent material strength in fatigue damage accumulation. To validate these findings, sensitivity analysis, PCC, and MIC were applied for comparative analysis. The results from these four analytical methods are mutually corroborative, forming a robust evidence chain that ensures the scientific validity and reliability of the weight analysis.



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Fig. 11. Analysis of factors influencing fatigue life.

It is noteworthy that factors such as loading frequency (f) and ambient temperature (T_e), which have rarely been quantitatively analyzed in traditional studies, were assigned definitive weight values in this investigation. This outcome underscores the unique

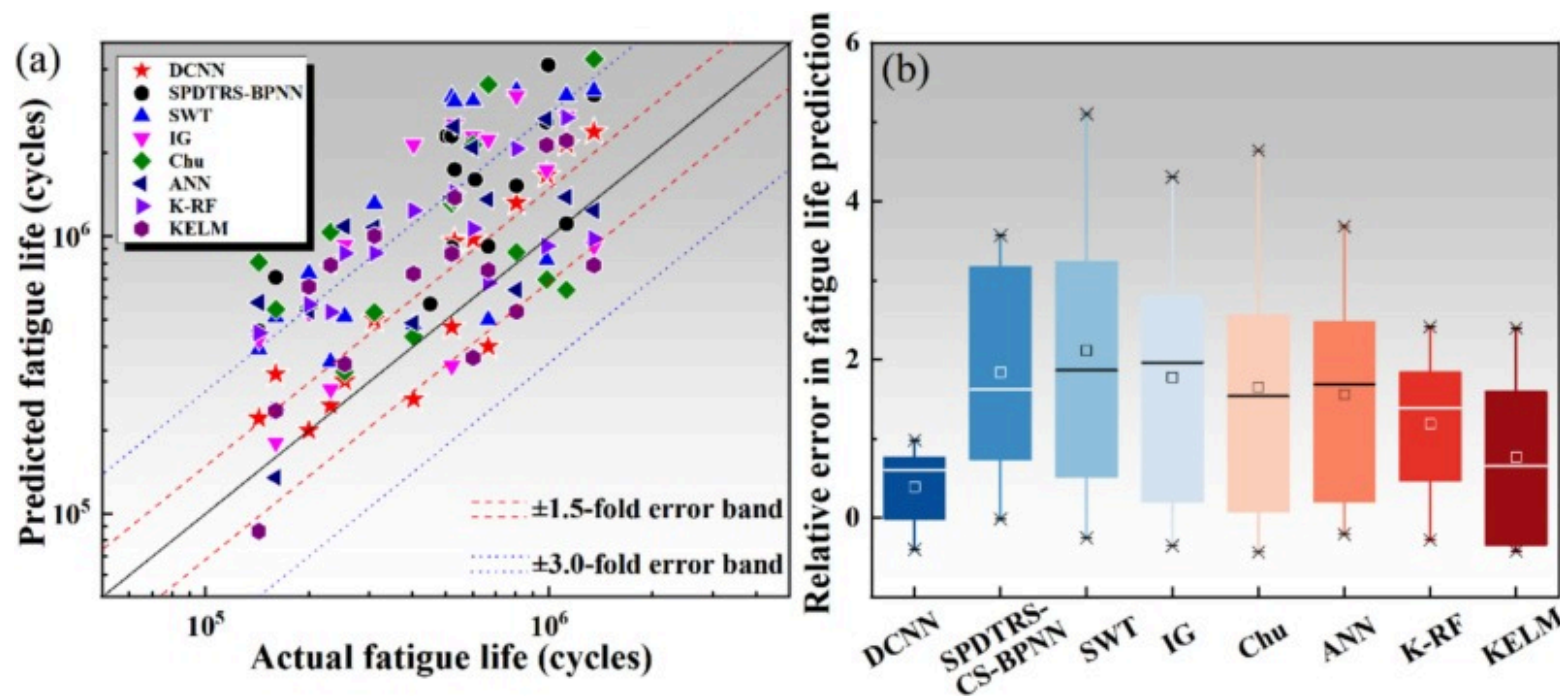
advantage of data-driven approaches in addressing complex multi-factor problems. Compared to the SPDTRS-CS-BPNN hybrid data-driven method, the DCNN-based approach coupled with influencing factor analysis and data augmentation incorporates a more comprehensive set of influencing factors. Nevertheless, the central importance of stress range and stress ratio remains unchanged, indicating their universal significance in fatigue life prediction.

5.3. Model predictive performance and comparative validation

To objectively evaluate the predictive performance of the proposed DCNN-IDA method, a systematic comparison was conducted against seven state-of-the-art fatigue life prediction models. These include the SPDTRS-CS-BPNN, SWT model, IG model, Chu model, ANN model, Kernel Random Forest (K-RF), and KELM model.

5.3.1. Prediction accuracy and error analysis

Fig. 12a compares the predicted and experimental results for all models. The predictions from the DCNN-IDA model are most closely distributed along the ideal prediction line ($y=x$), all falling within the ± 3 error band and primarily concentrated near the ± 1.5 error band. In contrast, the predictions from the other models show varying degrees of scatter. Notably, physical models like SWT and IG, as well as data-driven models like K-RF and KELM, exhibit obvious outlier predictions that exceed the ± 3 error band.



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Fig. 12. Comparison of different prediction models: (a) comparison of predicted and actual fatigue lives and (b) box plot of different prediction methods.

A boxplot analysis of the error distribution (Fig. 12b) quantitatively reveals the performance differences among the models. The DCNN-IDA model's boxplot features the smallest median, the narrowest interquartile range (IQR), and the smallest range, indicating superior prediction accuracy and stability. Specifically, the maximum relative error ($E_{\max}$) of the DCNN-IDA model is 0.98, which is 59.3% lower than that of the K-RF model (2.41) and 59.2% lower than the KELM model (2.40), demonstrating its robust performance even under extreme operating conditions.

5.3.2. Comparative analysis of comprehensive performance metrics

Table 6 summarizes the comprehensive performance metrics for all evaluated models. The proposed DCNN-IDA model achieves the best performance among all comparative models, with a MRE of 0.51 and a RMSE of 0.44. Specifically, its MRE shows reductions of 72.3% compared to the SPDTRS-CS-BPNN model, 76.5% compared to the SWT model, 72.7% compared to the IG model, and 47.4% compared to the KELM model. Furthermore, the model's coefficient of determination (R^2) reaches 0.927, providing additional evidence of its exceptional predictive capability.

Table 6. Comparison of accuracy and stability of different prediction methods.

	DCNN	SPDTRS-CS-BPNN	SWT	IG	Chu	ANN	K-RF	KELM
MRE	0.51	1.84	2.17	1.87	1.75	1.62	1.23	0.97
RMSE	0.44	1.19	1.66	1.40	1.57	1.28	0.84	0.97

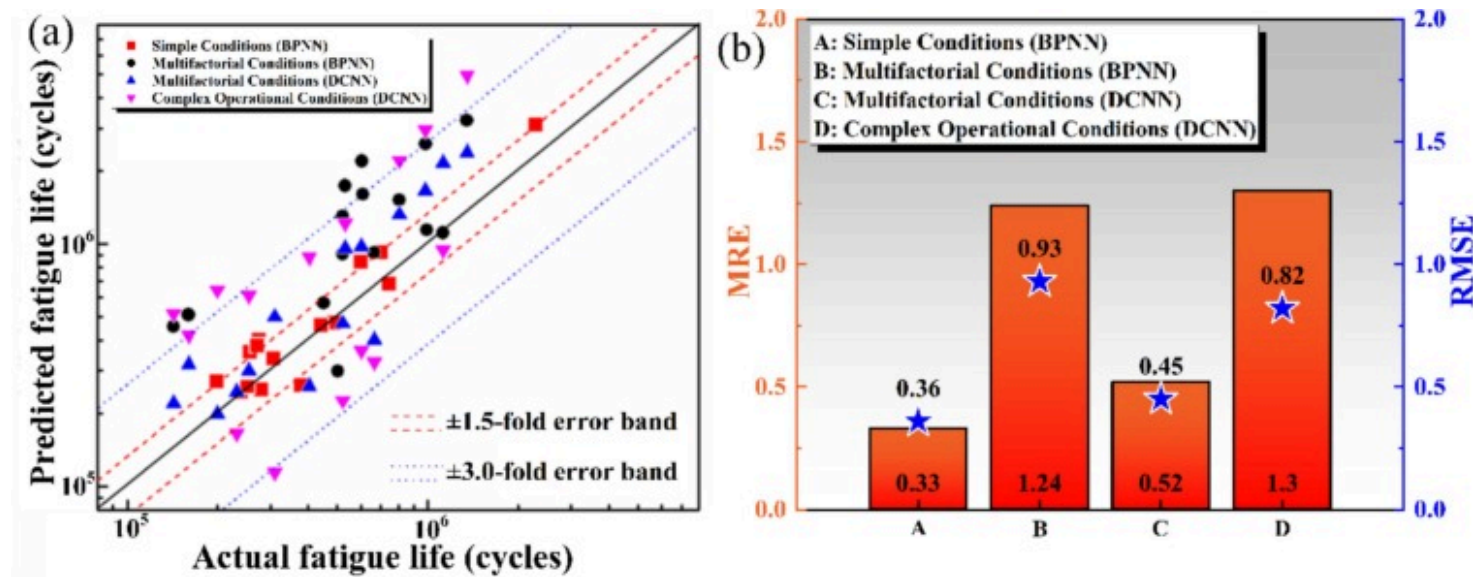
It is noteworthy that the traditional physics-driven models (SWT, IG, and Chu) demonstrate unsatisfactory performance under the multi-factor complex conditions investigated in this study, with MRE values all exceeding 1.75. This indicates inherent limitations of purely empirical formulations in addressing multi-field coupled fatigue problems. In contrast, while purely data-driven models (ANN, K-RF, and KELM) outperform the physics-based models, they still fall short of the performance achieved by the DCNN-IDA approach. This highlights the critical importance of enhanced data quality and interpretability guidance in constructing sophisticated predictive models for complex scenarios.

5.4. Discussion and limitations

Experimental results confirm that the DCNN-IDA method significantly improves the accuracy and reliability of fatigue life prediction under multi-factor varying conditions through the deep integration of data augmentation and interpretability analysis. The Borderline-SMOTE algorithm effectively mitigates data imbalance in complex feature spaces, while the synergistic application of XGBoost and explainable AI ensures the model's adherence to fundamental physical principles during learning.

However, it should be noted that the proposed approach remains fundamentally a data-driven paradigm. Although explainability tools offer valuable post-hoc insights, the model's learning process itself lacks strict constraints from physical laws. As shown in **Fig. 13**, prediction errors increase substantially when the model encounters extreme complex scenarios beyond the coverage of

training data. This indicates that in data-sparse regions, the model may produce predictions contradicting physical principles, thereby limiting its application in safety-critical engineering contexts.



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Fig. 13. Limitation analysis of fatigue life prediction methods: (a) error band comparison of prediction results and (b) comparison of prediction error.

This limitation reveals a fundamental weakness inherent in purely data-driven approaches: models operating without physical constraints carry systematic risks when generalized to untrained operational conditions. Consequently, moving beyond the current framework to explore a new generation of physics-informed data-driven methodologies—where physical governing equations are directly embedded into the model architecture—emerges as a critical and necessary direction for future research.

6. Fatigue life prediction method via physics-informed data-driven approach

While the methods proposed earlier have achieved significant progress in multi-factor prediction, purely data-driven approaches still exhibit inherent limitations in physical consistency and extrapolation capability. To address this challenge, this section

introduces a novel physics-data dual-driven method for predicting the fatigue life of welded joints. The proposed approach is founded upon physics-guided feature engineering and the innovative incorporation of physical principles into the neural network architecture. By embedding physical laws as constraints within the data-driven framework, this method ensures that fatigue life predictions remain consistent with the actual physical processes, thereby achieving reliable fatigue life estimation.

6.1. Formulation of physical constraints

The predictive fidelity of a data-driven model is inherently contingent upon the mechanical relevance of its input feature space. In the context of multivariable-coupled fatigue assessment, feeding raw, disjointed geometrical or operational parameters directly into a neural network often obscures the underlying physical mechanisms, thereby exacerbating the risk of non-physical mapping. To mitigate this, physics-guided feature engineering is deployed as the foundational cornerstone of the proposed dual-driven framework. This process explicitly translates abstract structural dimensions and loading variables into scientifically rigorous, mechanically meaningful indicators. By pre-conditioning the input space with established domain knowledge, this strategy not only elucidates complex multiphysics interactions but also fundamentally primes the network to capture the authentic driving forces behind fatigue damage evolution.

6.1.1. Physics-guided feature engineering for fatigue life influencing factors

In the fatigue life prediction of welded joints, weld geometry [55], [56], [57], [58], loading conditions [59], [60], and size effects [61], [62] are the primary determinants of fatigue performance. As substantiated by the importance analysis in the preceding sections, the local stress state and geometric configuration are critical to accurately assessing fatigue longevity. Consequently, the K_t is utilized as a core feature parameter, as it serves as a highly representative quantitative indicator that effectively couples these critical variables within geometrically discontinuous regions. Because stress concentration fundamentally accelerates the accumulation of fatigue damage, K_t directly dictates the local stress state and the underlying processes of crack initiation and propagation. Integrating K_t to preprocess the input data provides a physically grounded abstraction that facilitates the data-driven model's ability to internalize the intrinsic features of fatigue behavior. By transforming raw geometric dimensions into mechanically meaningful representations, this approach enables the model to more accurately capture the complex mappings between structural discontinuities and fatigue life, thereby significantly enhancing both prediction accuracy and stability.

It should be noted that characterizing local stress-strain responses in welded structures is inherently complex and typically demands extensive experimental testing. However, the primary focus of this research is the architectural integration of physics-

informed frameworks into machine learning models rather than the generation of primary experimental datasets. The K_t formulations employed in this work are based on established notch stress concentration theories and geometric parameter regressions sourced from published literature [63], [64]. By explicitly quantifying geometric discontinuities—specifically incorporating the weld toe radius (ρ), weld toe flank angle (α), and plate thickness (T)—an explicit computational system for K_t under varying joint configurations and stress fields has been established. For tension–compression loading, the governing equations are expressed as follows:

$$K_{t,\text{butt,tensile}} = 1 + \frac{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+2h+0.6h_p}{2h}}}{-0.45\pi\sqrt{\frac{T+2h+0.6h_p}{2h}}}\right)}}{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+2h+0.6h_p}{2h}}}{-0.45\pi\sqrt{\frac{T+2h+0.6h_p}{2h}}}\right)}} \cdot 2 \cdot \left[\frac{h}{\rho} \cdot \frac{1}{2.8\left(\frac{T+2h+0.6h_p}{T}\right)-2} \right]^{0.65} \quad (20)$$

$$K_{t,T,\text{tensile}} = 1 + \frac{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+2h+0.6(t_p+2h_p)}{2h}}}{-0.45\pi\sqrt{\frac{T+2h+0.6(t_p+2h_p)}{2h}}}\right)}}{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+2h+0.6(t_p+2h_p)}{2h}}}{-0.45\pi\sqrt{\frac{T+2h+0.6(t_p+2h_p)}{2h}}}\right)}} \cdot \left[\frac{h}{\rho} \cdot \frac{1}{2.8\left[\frac{T+2h+0.6(t_p+2h_p)}{T}\right]-2} \right]^{0.65} \quad (21)$$

$$K_{t,\text{cruciform,tensile}} = 1 + \frac{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+2h}{2h}}}{-0.45\pi\sqrt{\frac{T+2h}{2h}}}\right)}}{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+2h}{2h}}}{-0.45\pi\sqrt{\frac{T+2h}{2h}}}\right)}} \cdot \left[\frac{h}{\rho} \cdot \frac{1}{2.8\left(\frac{T+2h}{T}\right)-2} \right]^{0.467} \cdot [1.348 + 0.397 \ln\left(\frac{L}{T}\right)] \quad (22)$$

The corresponding calculation formulas under bending load conditions are provided in Eqs. (23) to (25).

$$K_{t,\text{butt,bending}} = 1 + \frac{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+2h+0.6h_p}{2h}}}{-0.45\pi\sqrt{\frac{T+2h+0.6h_p}{2h}}}\right)}}{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+2h+0.6h_p}{2h}}}{-0.45\pi\sqrt{\frac{T+2h+0.6h_p}{2h}}}\right)}} \cdot 1.5 \cdot \sqrt{\tanh\left(\frac{2t_p}{t+2h} + \frac{2\rho}{T}\right)} \cdot \tanh\left[\frac{\left(\frac{2h}{T}\right)^{0.25}}{1-\frac{\rho}{T}}\right] \cdot \frac{0.13+0.65\left(1-\frac{\rho}{T}\right)^4}{\left(\frac{\rho}{T}\right)^{\frac{1}{3}}} \quad (23)$$

$$K_{t,T,bending} = 1 + \frac{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+2h+0.3(t_p+2h_p)}{2h}}}{-0.45\pi\sqrt{\frac{T+2h+0.3(t_p+2h_p)}{2h}}}\right)}}{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+2h+0.3(t_p+2h_p)}{2h}}}{-0.45\pi\sqrt{\frac{T+2h+0.3(t_p+2h_p)}{2h}}}\right)}}} \cdot 1.9\sqrt{\tanh\left(\frac{2t_p}{T+2h} + \frac{2\rho}{T}\right)} \cdot \tanh\left[\frac{\left(\frac{2h}{T}\right)^{0.25}}{1-\frac{\rho}{T}}\right] \cdot \frac{0.13+0.65\left(1-\frac{\rho}{T}\right)^4}{\left(\frac{\rho}{T}\right)^{\frac{1}{3}}} \quad (24)$$

$$K_{t,cruciform,bending} = 1 + \frac{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+4h+0.3(t_p+2h_p)}{2h}}}{-0.45\pi\sqrt{\frac{T+4h+0.3(t_p+2h_p)}{2h}}}\right)}}{1-e^{\left(\frac{-0.9\alpha\sqrt{\frac{T+4h+0.3(t_p+2h_p)}{2h}}}{-0.45\pi\sqrt{\frac{T+4h+0.3(t_p+2h_p)}{2h}}}\right)}}} \cdot 1.9\sqrt{\tanh\left(\frac{2t_p}{T+2h} + \frac{2\rho}{T}\right)} \cdot \tanh\left[\frac{\left(\frac{2h}{T}\right)^{0.25}}{1-\frac{\rho}{T}}\right] \cdot \frac{0.13+0.65\left(1-\frac{\rho}{T}\right)^4}{\left(\frac{\rho}{T}\right)^{\frac{1}{3}}} \quad (25)$$

here, T represents the plate thickness, h is the reinforcement height of the butt weld, h_p denotes the weld width for butt welds and the leg length for T- and cruciform joints, L is the weld width for T and cruciform joints, t_p is the stiffener plate thickness in T- and cruciform joints, ρ is the weld toe radius (mm), and α is the weld toe angle ($^\circ$).

6.1.2. Physics-informed objective function

In current research, physics-driven models such as the Morrow, SWT, and IG models can, to some extent, account for the influence of mean stress on fatigue life, thereby enabling fatigue life prediction of welded joints. Therefore, the aforementioned physical models are utilized to constrain the data-driven approach. The corresponding computational process can be expressed by Eqs. (26) to (28):

$$\frac{\sigma_a}{E} + \left(\frac{\sigma_a}{H_c}\right)^{\frac{1}{h_c}} = \frac{\sigma'_f - \sigma_m}{E} (2N_a)^b + \varepsilon'_f (2N_a)^c \quad (26)$$

$$h_c = \frac{b}{c} \quad (27)$$

$$H_c = \frac{\sigma'_f}{(\varepsilon'_f)^{h_c}} \quad (28)$$

here, σ_m is the mean stress, E is the elastic modulus, σ_a is the stress amplitude, and σ'_f , $\sigma\epsilon'_f$, b , and c represent the fatigue strength coefficient, fatigue ductility coefficient, fatigue strength exponent, and fatigue ductility exponent, respectively. H_c and h_c denote the cyclic hardening coefficient and cyclic hardening exponent, respectively.

Furthermore, by adopting Eq. (29) as the objective function of the data-driven model, the integration of the introduced physical model and the data-driven approach is achieved. This enables the physical constraints of material strength, mean stress, weld geometry, loading conditions, and size effects to be incorporated at the objective function level. Through this integration, the developed fatigue life prediction model is able to learn the fatigue behavior of welded joints under complex loading conditions from the dataset, while ensuring that the prediction results comply with objective physical laws. Moreover, by physicalizing the objective function and optimizing the formulations involved in different processes, process control of the data-driven method under various fatigue conditions is realized, thereby enhancing the operability of the fatigue life prediction model.

$$\frac{\sigma_a}{E} + \left(\frac{\sigma_a}{H_c} \right)^{\frac{1}{h_c}} = \frac{(K_t \sigma_a)^2}{E \Delta \sigma} \quad (29)$$

Fatigue life prediction of welded joints is comprehensively affected by multiple non-linear factors. Consequently, standalone data-driven prediction models often produce severely deviating outliers or results that contradict objective physical laws, thereby undermining the model's reliability in engineering applications.

To address these limitations, the proposed dual-driven approach embeds physical laws into the network architecture by physically transforming the loss function. Specifically, rather than solely minimizing empirical data errors, the physical rules are formulated as penalty terms during the backpropagation process. This physically-guided optimization constrains the network's weight updates, ensuring the predictions remain physically plausible. As expressed in Eq. (30), this physics-informed loss function is composed of three distinct components ($Loss_{p1}$, $Loss_{p2}$, and $Loss_{p3}$), each carrying a clear physical meaning:

$$Loss_{\text{physics}} = Loss_{p1} + Loss_{p2} + Loss_{p3} \quad (30)$$

Generalization Across Working Conditions (Eq. 31, $Loss_{p1}$): This component captures the physical fatigue life discrepancies resulting from varying operational conditions among different samples. By penalizing unrealistic variations, it significantly enhances the model's ability to map complex physical input–output relationships.

$$Loss_{p1} = - \sum_{i=1}^{N_{total}} \sum_{j=1}^{N_{factor}} (y_{factor_{ij}} \ln p_{factor_{ij}}) \quad (31)$$

Non-Negative Life Constraint (Eq. 32, $Loss_{p2}$): This term strictly penalizes the network for predicting negative values, mathematically enforcing the fundamental physical reality that fatigue life must always satisfy $N_p \geq 0$. It eliminates the physically impossible predictions inherent in unconstrained data-driven models.

$$Loss_{p2} = \frac{\lambda_3}{N_{factor}} \cdot \sum_{i=1}^N ReLU(-N_{pi}) \quad (32)$$

Error Bound Constraint (Eq. 33, $Loss_{p3}$): Acknowledging the inherent statistical scatter and dispersion in physical fatigue testing, this constraint ensures that prediction errors are restricted within the objective bounds dictated by fatigue phenomena.

$$Loss_{p3} = \frac{\lambda_4}{N_{factor}} \cdot \sum_{i=1}^N ReLU(N_{pi} - \lambda_5) \quad (33)$$

where y_{factor} is a binary indicator variable (0 or 1); N_{total} denotes the total number of fatigue life influence factors within the current dataset; N_{factor} represents the cardinality (number of categories) of the influence factors; p_{factor} is the probability associated with each category; and N_p signifies the predicted fatigue life. The hyperparameters λ_3 , λ_4 , and λ_5 serve as penalty coefficients that dictate the relative importance of each respective physical constraint within the overall loss function [65]. It should be noted that unlike existing physics-informed studies that typically assign static, empirically predefined constants to these penalty terms, the proposed framework is specifically engineered for high applicability across intricate, multivariable-coupled operational conditions. Due to the severe disparities and high dispersion inherent in diverse engineering scenarios, attempting to assign a single, universal set of fixed values is fundamentally inadequate. Therefore, the basis for selecting these hyperparameters relies on a dynamically adaptive matrix rather than static numerical assignments. Their specific operational values are established through rigorous k-fold cross-validation during the training phase, uniquely tailored to the specific data distribution and boundary conditions of the target application. This adaptive tuning process ensures an optimal balance between the data-driven empirical loss and the physics-informed penalties, preventing any single constraint from dominating the gradient descent process and ensuring robust, stable model convergence across diverse structural assessments.

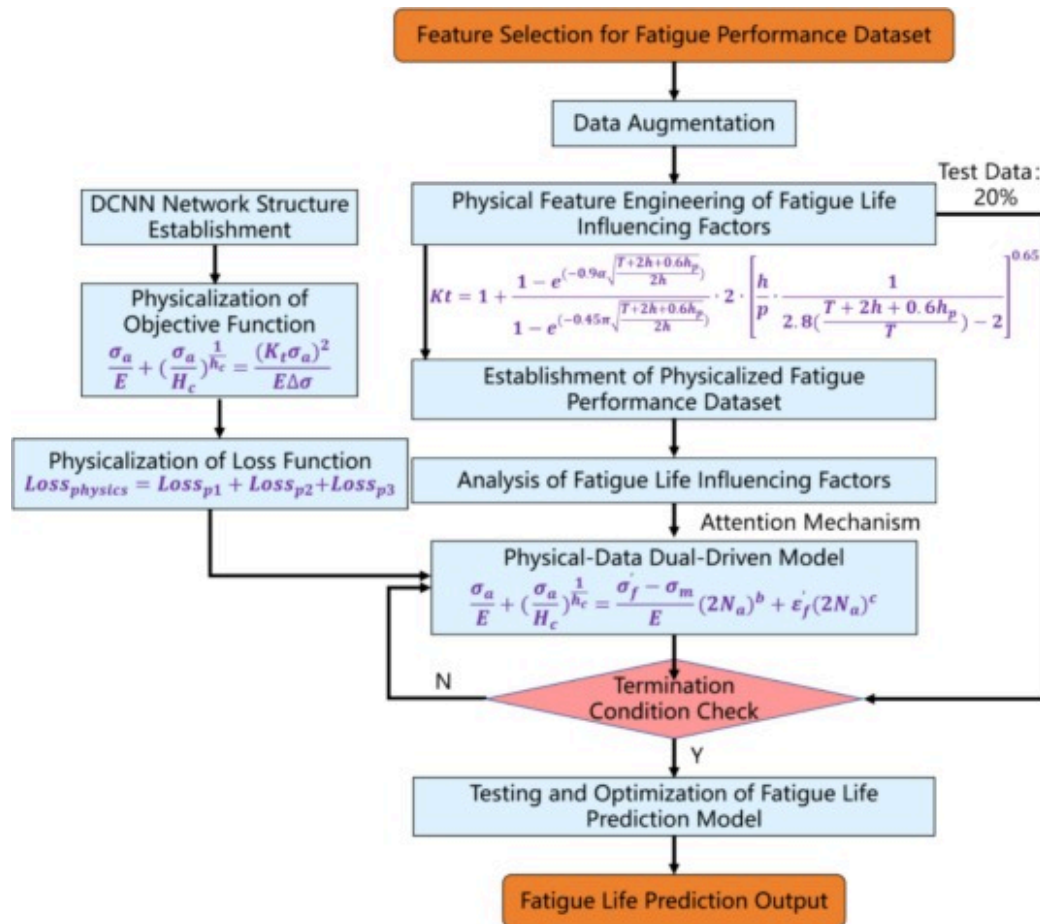
During the development of the physics-data dual-driven framework, the distinct components of the physics-informed loss function were explicitly tailored to address the unique physical challenges of fatigue life prediction. Specifically, $Loss_{p1}$ enables the model to accurately capture fatigue life discrepancies arising from varying operational conditions, thereby deepening the network's understanding of the complex mapping between influence factors and fatigue outcomes to improve overall accuracy and stability. Furthermore, $Loss_{p2}$ fundamentally enhances model reliability by preventing physically impossible negative predictions caused by potential network misinterpretations. Additionally, while high statistical scatter is intrinsic to fatigue phenomena, $Loss_{p3}$ effectively bounds the prediction errors within a reasonable margin, preventing excessive deviations that contradict objective physical laws.

Ultimately, by coupling physics-informed feature engineering with a physically constrained loss function, the proposed dual-driven approach ensures that all predictions strictly adhere to established physical laws, yielding a highly robust, accurate, and physically interpretable fatigue life prediction model.

To explicitly clarify the embedding mechanism and highlight its novelty, the physical constraints are integrated into the DCNN architecture as soft regularization terms rather than altering the forward-pass topology—which intrinsically avoids increasing computational overhead. Specifically, the overall embedding of physical knowledge is implemented through the optimization framework formulated in Eq. (26). Driven by a composite objective function (Eq. 29), the specific physics-informed loss components (Eq. 30 and its sub-components, Eqs. 31–33) act as penalty boundaries that regularize the standard data-driven empirical loss. Throughout the backpropagation phase, the gradients derived from this composite loss dynamically update the network's connection weights and biases. Mechanistically, this strategy physically restricts the mathematical hypothesis space during gradient descent. By actively penalizing predictions that contradict objective fatigue laws, it uniquely prevents the neural network from overfitting to localized noise and fundamentally eliminates the generation of physically impossible predictions, rigorously guiding the model to converge within physically consistent regimes.

6.2. Prediction framework and implementation

The physics-data dual-driven fatigue life prediction method achieves enhanced reliability by fully utilizing and effectively integrating both data-driven methodologies and physical laws, thereby establishing explicit physical constraints. The overall workflow is illustrated in Fig. 14 and can be described as follows:



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Fig. 14. Fatigue life prediction via physics-data driven method.

Step 1: Based on the established multi-scale fatigue performance dataset, the Borderline-SMOTE method is employed to perform data augmentation on the dataset containing physics-informed influencing factors. Taking the data augmentation process for the stress ratio feature as an example, the procedure can be described as follows:

Assume SR represents the complete fatigue performance dataset comprising all stress ratio features, where PR_{mi} denotes the values of stress ratios with relatively few instances in the dataset, and NR_{ma} denotes the values of stress ratios with relatively abundant instances. For each stress ratio value in set SR, the number of nearest neighbors (among m neighbors) that belong to the majority stress ratio class is denoted as m' (where $0 \leq m' \leq m$). Here, PR_{mi} and NR_{mi} can be expressed as:

$$P_{Rmi} = \{p_{R1}, p_{R2}, \dots, p_{Rn}\}, N_{Rma} = \{n_{R1}, n_{R2}, \dots, n_{Rn}\} \quad (34)$$

During the execution of the Borderline-SMOTE method:

- (1) If $m' = m$, the corresponding instance is considered noise.
- (2) If $m/2 \leq m' < m$, the instance is identified as being prone to misclassification and is thus categorized into the Danger Set (D_{Ra}).
- (3) If $0 \leq m' < m/2$, the instance p_s is considered to belong to the Safe Set.

Instances classified as noise or in the Safe Set require no further processing.

Based on this classification, the stress ratio features in set D_{Ra} (where $D_{Ra} \subseteq PR$) constitute the “Borderline” data. For each data group within D_{Ra} , its l nearest neighbors in set PR are identified. The set D_{Ra} is defined as:

$$D_{Ra} = \{p'_{R1}, p'_{R2}, \dots, p'_{Rd}\}, 0 \leq d \leq p \quad (35)$$

Subsequently, $e \times d$ synthetic instances are generated from the set D_{Ra} (where $1 < e < l$). For each instance p'_s in D_{Ra} , the differences (dif_c) between it and its e nearest neighbors from the minority class set PR_{mi} are calculated according to Eq. (36). Each difference dif_c is then multiplied by a random number r_c uniformly distributed between 0 and 1, thereby generating e new synthetic minority class instances:

$$synthetic_c = p'_s + r_c \times dif_j, c = 1, 2, \dots, e \quad (36)$$

Step 2: Process the fatigue performance dataset using the physics-guided feature engineering methodology for the influencing factors to create a physics-informed fatigue performance dataset, thereby providing a high-quality data foundation for model construction.

Step 3: Construct the DCNN architecture and optimize it using the physics-informed objective and loss functions.

Step 4: Perform an importance analysis of the fatigue life influencing factors using the XGBoost algorithm. Integrate these findings into the DCNN model via an attention mechanism, thus establishing the physics-data dual-driven fatigue life prediction framework. Simultaneously, validate the analytical results using PCC and MIC.

Step 5: Evaluate whether the termination criteria are met. If satisfied, conclude the training process and output the optimal solution; otherwise, proceed with the next iteration.

Step 6: Obtain fatigue life predictions by inputting the various influencing factors into the trained physics-data dual-driven prediction model.

6.3. Analysis of fatigue life influencing factors

Through physics-guided feature engineering and physics-informed function formulation, 14 physically-based influencing factors were derived. Among these, parameters including σ_m , E , σ'_f , ε'_f , b , c and σ_a are closely associated with the effects of material properties and mean stress on fatigue life, and are therefore defined as direct influencing factors. Parameters h , h_p , and t_p , which reflect the influence of loading conditions, are categorized into the dataset designated for analyzing loading effects. Furthermore, L , T , ρ , and α are utilized to characterize the effects of weld geometry and size on the fatigue life of welded joints.

As summarized in Table 7, four distinct datasets (Dataset 1 to Dataset 4) were constructed to facilitate appropriate analysis of the different types of influencing factors, with the corresponding data partitioning detailed for each.

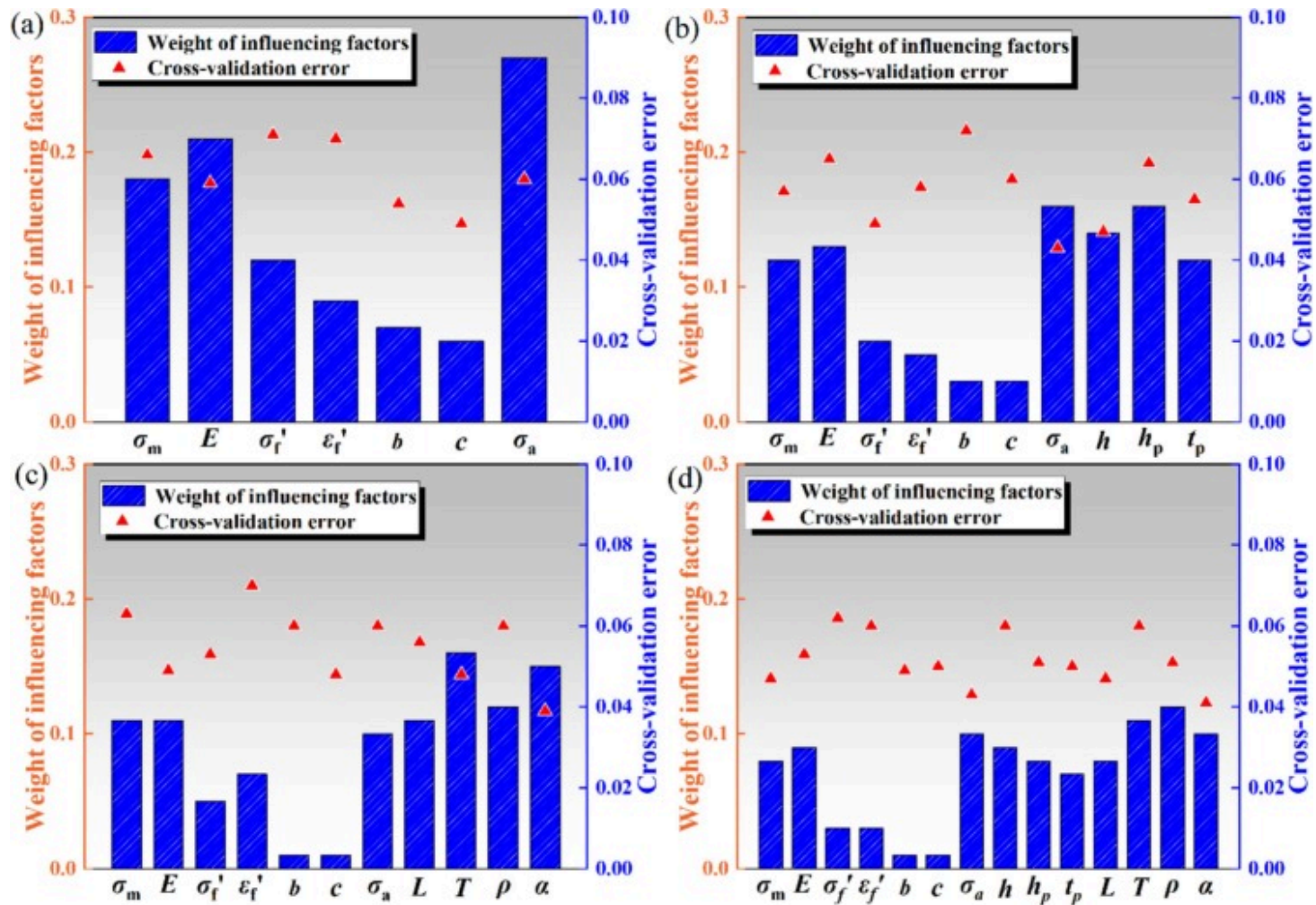
Table 7. Information of dataset partitioning.

Dataset Number	input	Output		Training set data volume	Test set data volume
		Direct prediction method	Indirect prediction method		
1	$\sigma_m, E, \sigma'_f, \varepsilon'_f, b, c, \sigma_a$	fatigue life	The parameters C and m in the S-N curve	935	231
2	$\sigma_m, E, \sigma'_f, \varepsilon'_f, b, c, \sigma_a, h, h_p, t_p$			800	179

Dataset Number	input	Output		Training set data volume	Test set data volume
		Direct prediction method	Indirect prediction method		
3	$\sigma_m, E, \sigma_f', \varepsilon_f', b, c, \sigma_a, L, T, \rho, \alpha$			810	233
4	$\sigma_m, E, \sigma_f', \varepsilon_f', b, c, \sigma_a, h, h_p, t_p, L, T, \rho, \alpha$			660	197

Using the XGBoost method, the influence weights of the factors within each of the four datasets on fatigue life were obtained. To comprehensively capture the varying degrees of influence exerted by different factors, a total of 14 physically-based influencing factors were considered. This approach aims to enhance model performance and broaden the applicability of the proposed factor analysis model and the fatigue life prediction framework. Furthermore, the weight analysis results were preliminarily validated using the cross-validation method as described in the scikit-learn library. The calculated weights are considered accurate when the relative error for each influencing factor is less than 10%.

A progressive, multi-level analysis was performed to systematically elucidate the influence mechanisms of key physical factors using four datasets of increasing complexity. As illustrated in Fig. 15, the relative importance of influencing factors evolves distinctly with dataset enrichment. Specifically, in Dataset 1, fatigue behavior is predominantly governed by the stress range, elastic modulus, and mean stress. With the inclusion of weld geometry parameters in Dataset 2, geometric features such as h , h_p , and t_p exhibit influence levels comparable to those of the fundamental mechanical factors. Further incorporation of size-effect descriptors in Dataset 3 leads to a pronounced increase in the contribution of parameters L , T , ρ , and α . Ultimately, the comprehensive evaluation based on Dataset 4 identifies the weld toe radius and plate thickness as the most critical determinants of fatigue life.



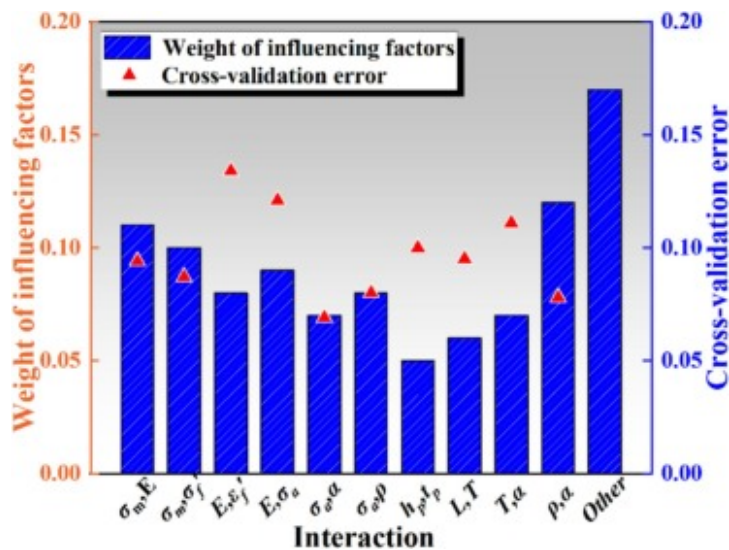
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Fig. 15. Weights analysis of influencing factors on fatigue life of welded joints under different datasets: (a) Set 1, (b) Set 2, (c) Set 3 and (d) Set 4.

This stepwise analysis clearly reveals the progressive evolution of factor importance under multi-physics coupling conditions, thereby providing a robust theoretical foundation for the construction of fatigue life prediction models.

In addition to single-factor contributions, the effects of inter-factor interactions were quantitatively assessed. Although individual interaction terms generally exhibit lower weights than dominant single factors, their cumulative influence is substantial and cannot be neglected. The interaction weight analysis presented in Fig. 16 and Table 8 highlights several key observations. Strong coupling is observed between material-related parameters, such as (σ_m, E) and (σ_m, σ'_f) , indicating intrinsic links between material deformation behavior and fatigue resistance. Load–geometry interactions, exemplified by (σ_a, ρ) , play a significant role in fatigue performance, underscoring the modulation of load effects by local stress concentrations. Moreover, interaction effects are found to be pervasive across nearly all factors, with their total contribution reaching a meaningful level that effectively transcends the limitations of conventional single-factor analyses.



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Fig. 16. Weight analysis of interactions of influencing factors on the fatigue life under Set 4.

Table 8. All weights of the interaction between influencing factors on fatigue life (%).

Influence factor	σ_m	E	σ'_f	ϵ'_f	b	c	σ_a	h	h_p	t_p	L	T	ρ	α
σ_m	8	11	10	4	2	2	1	2	2	3	1	1	4	3

Influence factor	σ_m	E	σ_f'	ε_f'	b	c	σ_a	h	h_p	t_p	L	T	ρ	α
E	11	9	4	8	3	3	9	1	2	1	1	1	1	2
σ_f'	10	4	3	4	1	2	4	2	1	1	2	3	1	2
ε_f'	4	8	4	3	1	1	3	2	2	1	2	4	2	2
b	2	3	1	1	1	1	1	0	0	0	1	2	1	2
c	2	3	2	1	1	1	2	0	1	1	1	1	3	4
σ_a	1	9	4	3	1	2	10	2	2	3	1	4	8	7
h	2	1	2	2	0	0	2	9	3	3	4	3	4	4
h_p	2	2	1	2	0	1	2	3	8	5	0	2	2	3
t_p	3	1	1	1	0	1	3	3	5	7	3	4	2	3
L	1	1	2	2	1	1	1	4	0	3	8	6	2	3
T	1	1	3	4	2	1	4	3	2	4	6	11	4	7
ρ	4	1	1	2	1	3	8	4	2	2	2	4	12	12
α	3	2	2	2	2	4	7	4	3	3	3	7	12	10

Collectively, these results offer a new theoretical perspective for interpreting fatigue behavior under complex service conditions characterized by strong multi-factor coupling.

7. Discussion

To comprehensively evaluate the robustness and predictive fidelity of the proposed methodology, this section details a rigorous comparative analysis against an array of established predictive paradigms.

7.1. Performance comparison of different fatigue life prediction models

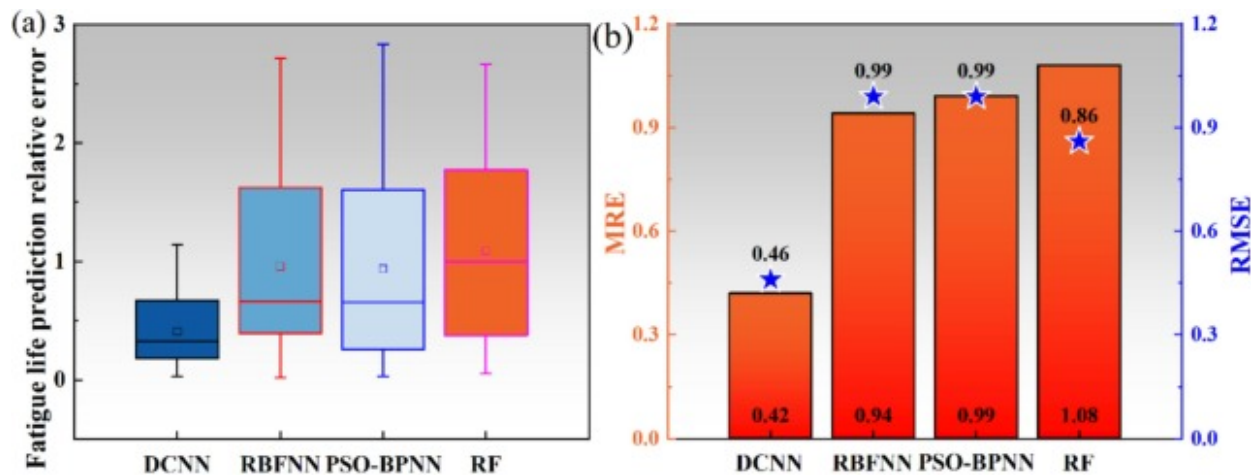
The benchmarking process is systematically divided into two complementary dimensions. The first phase evaluates the proposed model against contemporary purely data-driven algorithms, aiming to demonstrate the effectiveness of embedded physical constraints in mitigating overfitting and enhancing generalization within high-dimensional feature spaces. Subsequently, the second phase directly contrasts the framework with classical empirical fatigue models to substantiate its enhanced capacity for resolving complex, multivariable nonlinear interactions that traditional analytical methods frequently oversimplify.

7.1.1. Comparison with pure data-driven algorithms

The proposed physics-data dual-driven model achieved a maximum relative error of less than 1.0 across all validation cases, demonstrating significantly superior performance compared to the benchmark models. In contrast, predictions from the RBFNN, PSO-BPNN, and RF models mostly exhibited errors exceeding 1.0, with over 30% of their predictions even surpassing an error of 2.0. Further analysis indicates that the RBFNN model suffered from overfitting and limited generalization capability, the PSO-BPNN was prone to convergence to local optima, and the RF model struggled to capture complex interactions among high-dimensional features. These limitations stem primarily from an insufficient understanding of the underlying influencing factors and a lack of constraints from physical principles, ultimately restricting their predictive performance.

In comparison, the physics-data dual-driven model, by integrating physical laws with the DCNN architecture, effectively enhances the capability to comprehend complex, multi-faceted nonlinear relationships. This integration prevents physically implausible predictions while simultaneously mitigating issues of overfitting and local optima, thereby ensuring both the reliability and accuracy of the prediction outcomes.

As shown in [Fig. 17](#), the physics-data dual-driven DCNN model significantly outperforms other comparative models in terms of prediction accuracy. Statistically, its error distribution, represented by the boxplot, possesses the smallest interquartile range and the narrowest extreme value range ([Fig. 17a](#)). Quantitative analysis confirms that the DCNN model achieved the lowest MRE (0.42) and RMSE (0.46), representing reductions of 55.3% and 53.5% compared to the RBFNN model, 57.6% and 53.5% compared to the PSO-BPNN model, and 61.1% and 46.5% compared to the RF model, respectively ([Fig. 17b](#)). These results validate that the synergistic integration of physical constraints and the DCNN architecture effectively enhances the model's predictive accuracy and stability under complex operating conditions.



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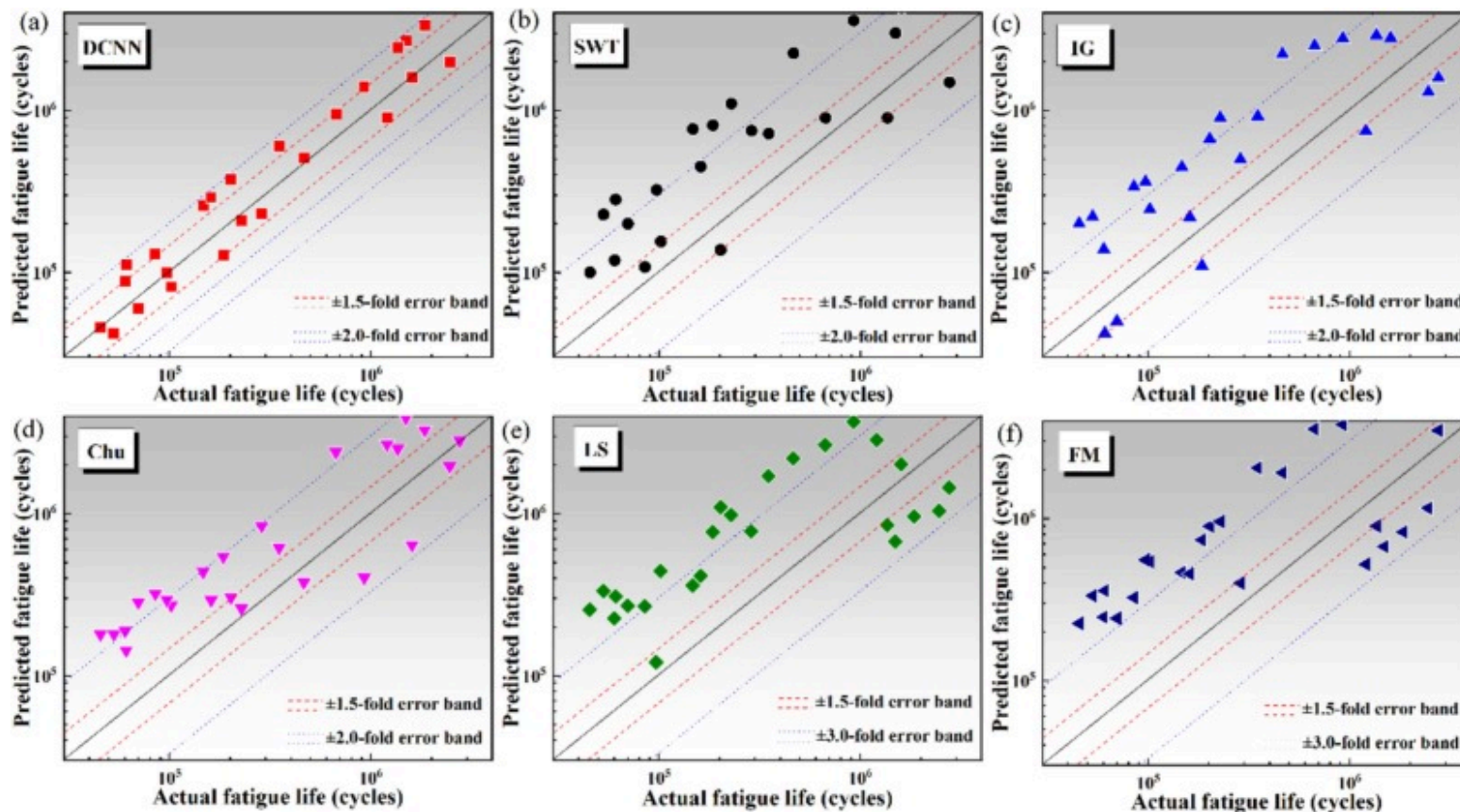
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Fig. 17. Performance comparison of different fatigue life prediction models under Set 4: (a) box-plot of fatigue life prediction error and (b) MRE and RMSE.

7.1.2. Comparison with traditional physical models

While the preceding comparative analysis robustly demonstrates the superiority of the physics-data dual-driven framework over purely data-driven algorithms—particularly in mitigating overfitting, avoiding local optima, and enhancing generalization within complex high-dimensional feature spaces—this alone does not fully elucidate the value of the embedded physical laws. Traditional empirical fatigue models inherently possess the physical rigor that pure machine learning models lack, yet they often struggle to comprehensively capture the non-linear, multi-factor coupling effects prevalent in actual engineering conditions. Therefore, to thoroughly validate both the physical plausibility and the architectural advantages of the proposed methodology, the following section systematically evaluates the dual-driven DCNN against classical physics-based models. Furthermore, an in-depth ablation study is conducted to unpack the underlying mechanism of the physics-informed loss function, revealing precisely how the integrated physical penalty terms guide the neural network to balance data-driven flexibility with objective mechanical constraints.

Traditional fatigue life prediction methods, including physics-based models such as SWT, IG, Chu, LS, and FM, have been widely used in practice. However, as shown in Fig. 18, the predictions of these conventional models exhibit significant scatter, with a large number of data points falling outside the ± 3 error band. Among them, the LS and FM models show poor applicability to welded joints, the SWT model demonstrates relatively better accuracy, while the FM model tends to yield conservative predictions. In contrast, the proposed physics-data dual-driven DCNN model demonstrates notable advantages, with its predictions evenly distributed within the ± 2 error band, indicating higher accuracy, smaller error fluctuations, and stronger adaptability to varying operating conditions.



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Fig. 18. Fatigue life prediction results via different models under Set 4: (a) DCNN model, (b) SWT model, (c) IG model, (d) Chu model, (e) LS model and (f) FM model.

As shown in Fig. 19, to explicitly evaluate the impact of physical constraints on model performance, a comparative analysis was conducted on the same dataset. The physics-data dual-driven DCNN model demonstrates a substantial improvement in predictive accuracy and stability over both the purely data-driven DCNN and traditional empirical models. Specifically, incorporating physical constraints reduces the MRE from 1.30 (standard DCNN) to 0.42, marking a 67.7% improvement. Similarly, the RMSE drops from 0.82 to 0.46. Fig. 19(a) further illustrates that the purely data-driven DCNN exhibits a wider dispersion in relative error, whereas the physics-guided architecture tightly constrains the error bounds, proving that physical laws effectively guide the neural network to avoid non-physical local optima.

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Fig. 19. Performance comparison of different fatigue life prediction models: (a) box-plot of fatigue life prediction error and (b) MRE and RMSE.

Furthermore, the proposed model outperforms classical fatigue evaluation methods significantly. The SWT model exhibits considerable errors (MRE: 1.97, RMSE: 1.41) due to its inability to account for complex multi-factor coupling effects. Although the IG and Chu models are based on the critical plane approach, they insufficiently consider material and joint-specific differences, with the IG model showing poor stability. The LS and FM models, being primarily designed for residual life assessment and difficult to adapt to multi-material characteristics, display the most pronounced error fluctuations and the highest overall errors (FM reaching an MRE of 2.67 and RMSE of 1.91). Overall, experimental results unequivocally confirm that the physics-data dual-driven method achieves optimal performance across all metrics, validating its superior effectiveness and reliability under complex operating conditions.

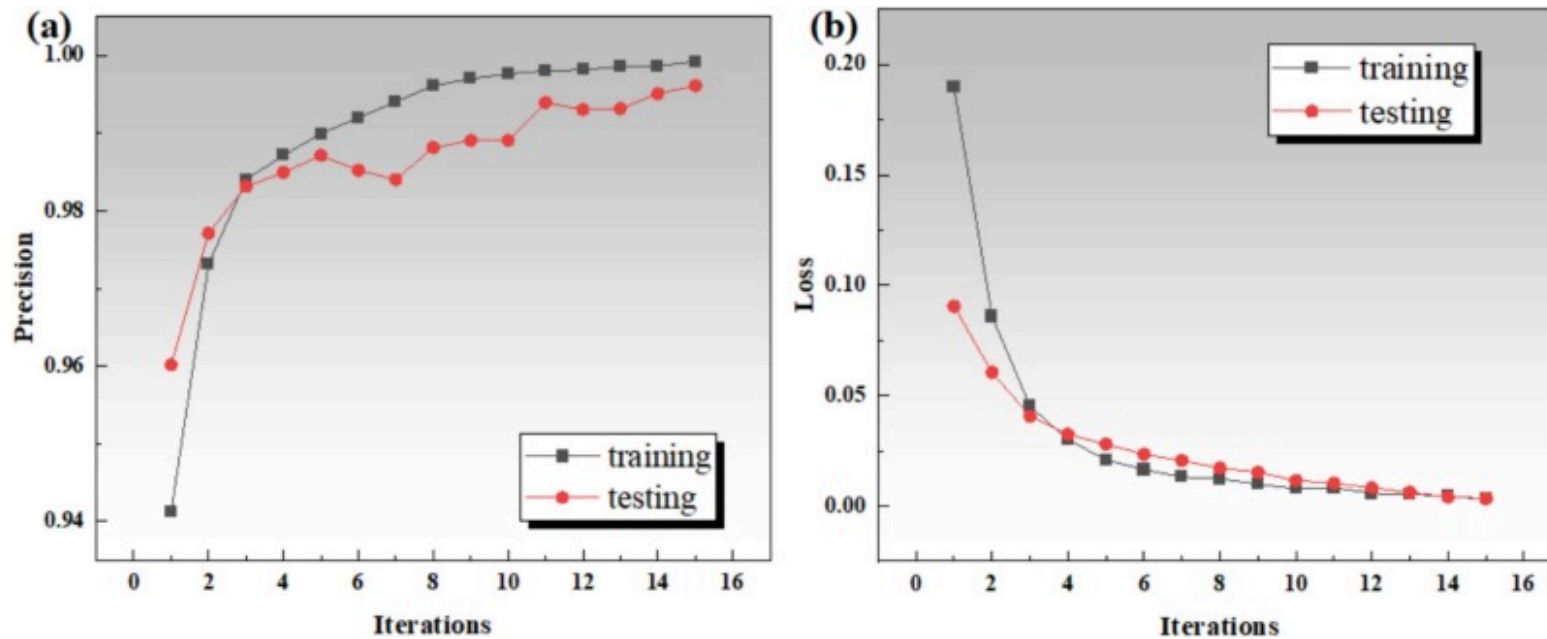
Mechanistically, the superiority of this physics-integration framework is inherently governed by the hyperparameters within the physics-informed loss function. A sensitivity analysis reveals the profound impact of these weights on overall model performance. When the weights approach zero, the framework degrades into a purely data-driven network; lacking macroscopic thermodynamic or fatigue damage accumulation boundaries, the hypothesis space becomes unconstrained, rendering the model

susceptible to high variance and localized noise. Conversely, excessively large weights over-constrain the neural network. From a physical standpoint, overly strict adherence to idealized macroscopic empirical formulas limits the network's capacity to autonomously learn the highly non-linear, microstructural, and complex multi-factor coupling damage mechanisms, ultimately resulting in underfitting. The optimal balance identified in this study successfully harmonizes the data-driven flexibility required to capture complex material characteristics with the rigorous adherence to fundamental physical laws.

7.2. Computational complexity and engineering practicality

To comprehensively address potential concerns regarding the computational complexity typically associated with deep neural networks, it is essential to highlight the inherent engineering practicality of the proposed dual-driven DCNN. Because the multi-scale fatigue performance dataset involves numerical structural factors rather than high-dimensional data, the network maintains a highly compact parameter scale. More critically, the integration of objective physical constraints restricts the mathematical hypothesis space, preventing the model from wandering into physically impossible local optima and thereby significantly accelerating the convergence process.

As illustrated in Fig. 20, during the training process, the DCNN model demonstrates extremely high efficiency. The prediction accuracy reaches 0.999 while the loss drops to a negligible 0.003 after merely 15 iterations. Concurrently, during the testing process, the accuracy stabilizes at 0.995 and the loss maintains at 0.003 within the same 15 iterations. The total training time required to complete this robust convergence is approximately 30s on a standard computational environment, yielding an overall R-squared (R^2) coefficient of 0.949. This conclusively proves that the physical constraints facilitate rapid learning with minimal computational overhead.



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Fig. 20. Performance index of the proposed model: (a) accuracy and (b) loss.

7.3. Prospects of data-driven fatigue life prediction methods for welded structures

The findings of this study underscore that the efficacy of a data-driven fatigue life prediction model is highly sensitive to the comprehensiveness of the considered influencing factors. Welded joint fatigue is governed by highly nonlinear, multivariable interactions intricately tied to varying environmental conditions and manufacturing fluctuations. Given the current absence of sufficiently massive fatigue performance databases, aggressive dimensionality reduction—a standard practice in conventional data-driven workflows—may prove counterproductive. Retaining a broader spectrum of physical features is crucial for enabling algorithms to capture the authentic underlying mechanics. However, compiling such high-fidelity, multi-scale datasets is exceptionally time-consuming and resource-intensive. Consequently, a concerted, collaborative effort across the global research community is imperative to establish open-access, standardized fatigue performance databases, which will serve as the critical catalyst for advancing next-generation prediction frameworks.

Furthermore, the constraints imposed by current data volumes dictate a judicious approach to model architecture. While deep learning algorithms possess formidable capacity for mapping complex fatigue behaviors, blindly pursuing excessive architectural complexity is unwarranted. The principle of parsimony should guide model selection, prioritizing architectures tailored to the specific complexity of the target problem. In scenarios constrained by limited data, a well-optimized, simpler network often outperforms an insufficiently trained deep neural network (DNN). For instance, while the proposed SPDTRS-CS-BPNN hybrid model exhibits limitations under extreme multivariable-coupled conditions, it provides a highly efficient and accurate predictive solution for simpler standardized test specimens.

Ultimately, the maturation of reliable data-driven fatigue life prediction for welded structures relies on a triad of foundational pillars: a robust empirical database serving as the foundation, an algorithm explicitly tailored to the specific engineering context, and the rigorous integration of objective physical laws to constrain empirical learning. Through iterative training, experimental validation, and architectural optimization, highly robust predictive paradigms can be systematically established.

Beyond the immediate predictive enhancements, the methodologies developed herein hold substantial promise for broader engineering applications. By rigorously enforcing physics-informed constraints, the framework guarantees prognostic reliability even under uncharted operational regimes, while its interpretability analysis provides a transparent, quantifiable basis for safety assessments. Furthermore, the modular architecture of this dual-driven paradigm enables seamless integration into existing design specifications and advanced structural health monitoring (SHM) systems. Future research endeavors will pivot towards the physics-informed modeling of multi-field coupled fatigue, the formulation of transfer learning strategies for data-sparse scenarios, and the optimization of dynamic life prediction algorithms utilizing real-time monitoring streams. Ultimately, these targeted advancements are anticipated to catalyze the widespread industrial deployment of physics-guided artificial intelligence in structural integrity management.

8. Conclusion

This study proposes a novel physics-informed data-driven framework for the fatigue life prediction of engineering welded structures, successfully bridging the gap between empirical damage mechanics and deep learning. The primary conclusions are as follows:

- (1) A hybrid SPDTRS-CS-BPNN model was developed for baseline prediction. By utilizing closed-loop optimization for precise feature weighting and parameter tuning, the model effectively suppressed overfitting and prevented local optima, achieving exceptional accuracy with a Mean Relative Error of 0.35 and a Root Mean Square Error of 0.38.
- (2) Feature analysis revealed that while stress range dictates fatigue life under simple loading, synergistic couplings of joint geometry, size effects, and material properties dominate under complex conditions. The proposed deep learning framework, integrating data augmentation and attention mechanisms, successfully captured these multivariable interactions and reduced prediction errors by over 45%.
- (3) A physics-data dual-driven architecture was established by explicitly embedding objective physical laws, such as non-negative life reality and mean stress corrections, as dynamic penalty terms directly into the loss function. This strict physical regularization effectively eliminates physically implausible predictions, drastically reducing errors in data-sparse regions by 42.7% and enhancing extrapolation stability by 35.3%.
- (4) Benefiting from physics-guided feature engineering, the dual-driven framework demonstrated exceptional robustness against the inherent dispersion of fatigue data. It achieved a mean relative error of 0.42 and kept maximum relative errors strictly below 1.0, reliably outperforming traditional empirical methods and unconstrained algorithms by over 50%.

CRedit authorship contribution statement

Yuhang Duan: Writing – original draft, Formal analysis. **Chao Feng:** Writing – review & editing, Funding acquisition, Conceptualization. **Lianyong Xu:** Project administration, Methodology. **Lei Zhao:** Visualization, Validation. **Xiaohong Zhan:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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